



Supply chain product visibility: Methods, systems and impacts



Ahmed Musa^{a,*}, Angappa Gunasekaran^b, Yahaya Yusuf^a

^a Division of Systems and Operations, Lancashire Business School, University of Central Lancashire, Preston PR1 2HE, United Kingdom

^b Department of Decision and Information Sciences, Charlton College of Business, University of Massachusetts–Dartmouth, North Dartmouth, MA, United States

ARTICLE INFO

Keywords:

RFID
Tracking and tracing
Sensor networks
Architecture design

ABSTRACT

Supply chain product visibility may be defined to mean the capacity of the supply chain to have a view of a product's lifecycle, from its conception, manufacturing, distribution, delivery to the end customer, customer's experience of the product, and the product's end-of-life activities and processes. This implies developing and keeping a record of the product's materials and components, its physical state throughout the supply chain, the product's forward movement to the user-customer, customer's experience of the product, and the reverse logistics and reuse or termination of the product. The aim of visibility is to foster planning, control and agility of operations associated with the product and to improve customer experience of the product. "Tracking" is the term often used to describe the determination of the identity and state of a product in the forward direction (from manufacturing to the end user), while "tracing" is used to infer the product's path and history from downstream to upstream of the supply chain. In recent times there has been an upsurge of academic and commercial interest in product visibility. This interest has translated into numerous architectures, technologies and software for product visibility, both at the atomic (item) and composite (or aggregate) levels. Based on an extensive content analysis of academic and trade literature, including websites and documents of vendors and users of the technologies, this paper captures, analyses, compares and contrasts the design choices, essence, results and current and potential future impacts of some of the recent developments. The study also used survey by questionnaire across industries to assess user requirements of tracking and tracing systems and structures. The paper also charts future research directions for end-to-end visibility of product classes and their instances in supply chains.

Crown Copyright © 2013 Published by Elsevier Ltd. All rights reserved.

1. Introduction

End-to-end supply chain product visibility by product tracking and tracing has been exploited as a means of product security and process control and optimization in many industrial sectors, ranging from manufacturing, transportation and distribution, retailing, aviation, healthcare (including management of pharmaceuticals and patients), agriculture to food safety (Bottani & Rizzi, 2008; Hsu, Levermore, Carothers, & Babin, 2007; Hsu & Liao, 2011; Hsu & Wallace, 2007; Lee & Lee, 2010, 2012; Lee & Özer, 2007; MAFF Japan, 2003). Ongoing rapid developments in RFID, the evolution of communication and localization technologies (such as XML, ebXML, EDI, Bluetooth, WiFi, WiMax, WiBro, Zigbee, Ultra-Wide Band, RuBee, IEEE 802 family of standards, infrared, indoor messaging), MEMS-based small sensors and actuators, web services, multi-agents, together with the digitization of public infrastructures in the era of "internet of things" (IoT) have ushered in several methods, systems and architectures for achieving visibil-

ity of products classes and instances across supply chains. Product visibility is achieved by tracking and tracing the product, possibly throughout its lifecycle, using a variety of methods and technologies, like barcode, RFID, communication channels and sensor networks. Supply chain tracking is the ability to follow the design, composition, processing, application, location and forward path of a product or batch/lot downstream of the supply chain to the end customer. Tracing, on the other hand, is the ability to ascertain the origin, path, history, design, composition, etc., of the product upstream the supply chain. Tracking and tracing are complementary activities and their aims are to increase the security of the product, streamline and optimize production planning and distribution systems and processes, locate sources of quality issues (faults or contamination), manage recalls efficiently, etc.

In the food industry, visibility has witnessed considerable interest and attention in recent years (Amador & Emond, 2010; Bechini, Cimino, Marcelloni, & Tomasi, 2008; Bechini, Cimino, & Tomasi, 2005; Trienekens, Wognum, van der Beulens, & Vorst, 2012). The reasons for the interest in RFID by the food industry include the fact that food visibility has become a legal obligation in many jurisdictions and companies in the food supply chain now consider investment in visibility as a strategic move to promote public

* Corresponding author. Tel.: +44 1772894621; fax: +44 1772892906.

E-mail addresses: amusa@uclan.ac.uk (A. Musa), agunasekaran@umassd.edu (A. Gunasekaran), yyusuf@uclan.ac.uk (Y. Yusuf).

confidence in their products. The need for and benefits of visibility have also been suggested in the context of the virtual enterprise, supply chain risk management, agent-based control of industrial systems, and strategic and dynamic business reconfiguration and alignment. Visibility also plays an important role in closed-loop product lifecycle information management (PLIM) (Kiritsis, 2011; Ranasinghe, Harrison, Främling, & McFarlane, 2011). The airline industry is making a faster progress in the adoption of PLIM (Airlines of America, 2009, chap. 9) but there are pending challenges.

To date, there exist many RFID and sensor network-based architectures and hardware for supply chain product visibility. Some have been simply proposed and possibly prototyped in the literature while a few others have actually been deployed by the industry. They do not always satisfy the same set of requirements and their applicability is domain-specific. Therefore, it is relevant to seek to determine and characterize the application domains as well as the similarities, differences, advantages and disadvantages between the different network, hardware and middleware configurations and designs. To the knowledge of the authors, there is no up-to-date literature that comprehensively classifies and differentiates between the currently available visibility configurations and systems. This paper fills this gap. The classification and comparisons by Främling, Ala-Risku, Kärkkäinen, and Holmström (2007) and Främling, Harrison, Brusey, and Petrow (2007) appears incomplete, being limited to three systems and to object identifier formats; see also Holmström, Kajosaari, Främling, and Langius (2009). Främling et al. (2007) and Främling and Harrison et al. (2007) compares only EPC network, DIALOG (ID@URI) and Worldwide Article Information (WWAI) systems. The work by Ranasinghe et al. (2011) compare the same systems as in Främling et al. (2007) and Främling and Harrison et al. (2007) and examined their suitability to the specific needs of product lifecycle information management (PLIM) of three case study companies (Fiat, Caterpillar, and Indesit). The criteria used by this paper to classify and distinguish between systems for supply chain product visibility were derived from actual user requirements. The user needs were determined by conducting a survey by questionnaire.

The paper is organized as follows. Sections 2 and 3, respectively, present the background to the study and the research methodology. Section 4 discusses the methods for supply chain product visibility, and Section 5 presents hardware and software configurations for product visibility, the properties of visibility systems, as well as how the properties vary amongst the various systems. Section 6 presents the results of a survey by questionnaire on user requirements of visibility systems. The requirements emerged from extensive literature review and consultations with practitioners via questionnaire survey and interviews. Section 7 discusses the existing architectures and structures for supply chain visibility, and Section 8 compares four of the frameworks for visibility on the basis of a sub-list of user requirements of Section 6. Because it is not possible to describe and discuss all the structures to great length in this paper, only four of them are described in detail. Those four were selected because they are currently in deployment in the industry and are not simply proposals. Section 9 concludes the paper.

2. Background to the study

Some analytical, simulation and empirical studies have shown that racking can reduce retailer inventory costs by up to 30% (Hong, Kim, & Kim, 2010; Hong, Yoo, Ko, & Kim, 2008; Lee & Özer, 2007), while others have estimated savings of at most only 2% (Ferrer, Dew, & Apte, 2010; Sarac, Absi, & Dauzère-Pérez, 2010; Shafiei, Sundaram, & Piramuthu, 2012; Souza et al., 2011). For a number of different motives, in the food industry, visibility

has seen a significant interest in recent years (Amador & Edmond, 2010; Bechini et al., 2005, 2008). The reasons include the fact that food visibility has become a legal obligation in many jurisdictions (FDA, 2004; Food Standards Agency, 2002a, 2002b) and companies in the food supply chain now consider investment in visibility as a strategic asset to promote public confidence in their products. The need for and benefits of visibility have also been suggested in the context of the virtual enterprise (Bechini et al., 2005, 2008), the internet of things (Atzori, Iera, & Morabito, 2010), quality assurance beyond the food supply chain (Tse & Tan, 2012; Xu, 2011), supply chain risk management (Gomez, Laurent, & El Moustaine, 2012), agent-based control of industrial systems (Hsu et al., 2007; Hsu & Wallace, 2007; Vrba, Macůrek, & Mařík, 2008), and strategic and dynamic business reconfiguration and alignment with the marketplace (Chen, Cheng, & Huang, 2013; Jedermann & Lang, 2008; Véronneau & Roy, 2009; White, Daniel, & Mohdzain, 2005; Xia & Chen, 2011).

The increasing feasibility of peer-to-peer (P2P) communications, as often used in electronic data interchange (EDI), decreases the need for centralized exchanges and makes electronic dyads more attractive (Bechini et al., 2008). However, it is also argued that concepts like RFID, web services (WS), electronic business using eXtensible Markup Language (eXML), Enterprise Service Bus (ESB), and Service-Oriented Architecture (SOA) hold better prospects for enabling product visibility and dynamic collaborative environments and business process integration (MIMOSA, 2006). In data exchange, RFID plays the essential role of automatically identifying mobile products and/or lots (Ahn, Childerhouse, Vossen, & Lee, 2012; Chappel, 2004; Chatziantoniou, Pramatarí, & Sotiropoulos, 2011; Gibbs & Damodaran, 2002; Zhou, 2009).

Visibility also plays an important role in closed-loop product lifecycle information management (PLIM), a concept whose adoption and realization by the industry has been understandably slow and incomplete (Cassina, Taisch, Potter, & Parlikad, 2008; Främling, Ala-Risku, Kärkkäinen, & Holmström, 2006; Främling et al., 2007; Främling & Harrison et al., 2007; Kiritsis, 2011; Ranasinghe et al., 2011). PLIM is being pursued more vigorously by the Airline industry than other industries (Airlines of America, 2009, chap. 9) but there are still challenges outstanding. PLIM is composed of three interrelated or intertwined phases: beginning of life (BoL), middle of life (MoL), and end of life (EoL). Product conceptualization and manufacturing constitute the BoL phase, while the MoL stage begins as the product leaves the retailer or dealership. The EoL stage begins when the product reaches the end of its usage by the end user. During EoL, the product may be reprocessed for reuse or resale, recycled or disposed/discarded. Drivers like environmental regulations, cost recovery, safety and warranty management, and greenwashing (i.e., the scenario of a company cultivating a positive public perception of its green credentials while not investing enough in greening the supply chain) are increasingly motivating improved and wider application of EoL strategies. In some industries, e.g., the aircraft or heavy industry, the MoL phase may last for up to 30 years. PLIM has the following potentials: assist manufacturers in making product design improvements using data obtained from PLIM; using product lifecycle to innovate and develop new product lines; adapting production and distribution systems to improve product design, performance and efficiency; developing the capacity for predictive maintenance so as to reduce or eliminate the need for unscheduled maintenance; forecasting possible product failures; providing instant access to maintenance and usage histories of products and components; assessing the suitability of a product or component for recycling, reuse, or remanufacturing; and deciding any legal or trade regulations concerning the recycling or disposal of the product or its parts.

Visibility is a complex task. The complexity results from several factors, the main ones of which include: the fact that visibility data

is often dispersed over different enterprises, geographical locations and information systems; the need for efficient traceability to cope with challenges such as lack of agreement of potentially disparate legacy enterprise information and communication systems; the inhomogeneity of distributed databases; global sourcing and manufacturing and the resulting spread and complexity of modern supply chains; visibility data may be lot-based, product-class or stock keeping unit (SKU) based for some products and parts while being item-based for some other products or their sub-assemblies (Aberdeen Group., 2011; Agaidi., 2010; Bottani, Montanari, & Volpi, 2010; Sarac et al., 2010; Tajima, 2007; Véronneau & Roy, 2009). The building of supply chain visibility platforms has also to contend with problems generally associated with justifying information technologies/systems (IT/IS): absence of IT-related performance measures that take into account organizational features as well as strategic, tactical and operational objectives; assessment and management of risks associated with IT implementation; lack of optimization techniques for IT evaluation and selection and for IT investment choices (Akyuz & Rehan, 2009; Gunasekaran & Ngai, 2004; Gunasekaran, Ngai, & McGaughey, 2006; Tse & Tan, 2012).

The apparent complexity of visibility is aptly demonstrated by Xu, Wijesooriya, Wang, and Beydoun (2011) who presented an outbound logistics exception monitoring approach by incorporating multi-perspective ontologies and intelligent agents. The multi-perspective ontologies, involving static ontology, social ontology and dynamic ontology, are used to reflect the dependencies of logistics exception and to represent the dynamic nature of business processes. Xu et al. (2011) then designed an outbound logistics exception monitoring system by introducing intelligent multi-agents that can enforce autonomous, flexible, and collaborative exception monitoring. The case studies presented by Xu et al. (2011) indicated that multi-perspective ontologies provide a more accurate and realistic appreciation and modeling of exceptions.

After a comprehensive survey of the literature we identified several research gaps in supply chain visibility. One of such gaps is that, while there are several systems and software and hardware architectures for product visibility, the user requirements for these systems have not been adequately captured by extant literature. Another area of research is the need for systematic domain-wise evaluation and comparison of existing systems and architectures for visibility. Such studies will assist potential users in identify systems that are best suited to their needs as well as enable researchers to determine the interoperability requirements for such systems. It will also help in the development of a system or a group of systems that combine the best features of existing systems while offsetting their deficiencies. This paper contributes to filling the research gaps of user requirements for product visibility systems and the comparison of existing product visibility architectural frameworks.

Another area of research identified from the literature review is closed-loop product lifecycle information management (PLIM). The airline industry is making a faster progress in the adoption of PLIM (Airlines of America, 2009, chap. 9), but the fact is that even within the aviation industry the realization of the full potential of this concept has been logically very gradual. Some of the challenges to industry-wide adoption of PLIM were identified by the PROMISE project (PROMISE, 2008). Currently, the main challenges are three: the cost of sufficiently powerful sensors and embedded devices; the need for interoperability of different systems and devices for exchange of static and dynamic data; and user attitude towards product's end-of-life needs (e.g., disposition, reuse and/or recycling instructions). The power of the embedded devices is interpreted in terms of the capacities required for sensing, memory, onboard processing, communication and battery. Processing power (and hence battery power) is needed for communication, for running algorithms for

product residual life management, telematics activities by the micro-device (data aggregation, update, etc.), onboard database management, decision making, and other aspects of decision support. Ontology-based approaches for semantic standards are a likely solution to seamless interoperability of disparate systems.

A third area of current need is the linking of product visibility with the processes, activities and actors to which the product is subjected in its lifecycle. PLIM holds the prospects to enabling this functionality.

As mentioned earlier in this section, this paper makes a contribution to the understanding of user requirements for product visibility systems and of the similarities and differences between the hardware and software architectures of existing product visibility systems.

3. Research objectives and approach

At each node of the supply chain the components of a product visibility system may be configured uniquely. For a given enterprise or supply chain, the standard and mechanism by which the visibility systems at different nodes communicate and interact with each other may also exhibit some peculiar features. The aim of this paper is to characterize existing RFID and sensor-based product visibility hardware and deployment frameworks, as well as study the similarities and differences between the different network, hardware and middleware configurations and designs. In recent years many commercial and academic RFID and sensor-based product visibility hardware and deployment frameworks have been introduced, and, given the rate of activity in this area, new ones are continually being introduced. Academic/research deployment architectural frameworks and systems have been simply proposed and trialled in the literature. On the other hand, the commercial systems are actually in use in the industry. The current products and deployment frameworks do not always satisfy the same requirements and their applicability is often domain-specific. There does not seem to exist an up-to-date literature that comprehensively classifies and distinguishes between the currently available visibility configurations and systems. This paper removes this chasm by determining and characterizing the application domains as well as the similarities and differences between the different hardware and middleware configurations and schemes.

Also, our extensive survey of the literature has indicated that not much research has been conducted to characterize end-user requirements for product visibility systems. In particular, since the existing visibility architectures and systems have different features and fulfil different design and operational criteria, it is pertinent to inquire into the classes of features that are most relevant to various application scenarios and markets.

This paper plugs these knowledge gaps by formulating and answering the following research questions:

- (a) How can the generic term “product visibility” be described and modeled with academic rigour and what forms of visibility exist?
- (b) Which product visibility standards, systems and structures exist in the academic and commercial arenas?
- (c) What are the parameters, dimensions and features that can be used to distinguish between the various technologies and standards that exist for supply chain visibility?
- (d) What are the significant differences between existing standards, architectures and products for visibility in terms of some essential features and properties and user needs?
- (e) What challenges exist in unifying and standardizing the current designs and architectures? Standardization does lead to wider technology acceptance and price reduction.

In answering these questions this paper adopted the following research methodology. The main research instruments used by the papers were literature review and synthesis as well as survey by questionnaire. We used academic and trade literature to determine the characteristics of current visibility systems and products and the user requirements that those systems should satisfy. For academic sources, we used Discovery/EBSCOhost electronic databases for journals that are subscribed to by the University of Central Lancashire library. The search strings used were “RFID OR visibility AND supply chain”, “tracking OR tracing AND supply chain”, and “intelligent product OR object”. In total, more than 400 academic articles were selected and studied. Our interest was mainly in sources that presented materials related to technologies for supply chain visibility, intelligent products, and their direct application to supply chain management. A second selection process led to the elimination of more than 300 articles, leaving only 100 academic articles for the review that led to this paper. We also surveyed the websites and documents of vendors and users of RFID and visibility technologies.

The second aspect of the research methodology was the use of survey by questionnaire to assess the industry's perception of the requirements for visibility systems, i.e., the features of such systems that are considered by practitioners to be most relevant. We developed from the extensive literature review a set of requirements which the questionnaire respondents were simply asked to score on a scale of 1 to 10, as a measure of the importance they would accord the features if they have adopted or were to adopt a visibility system. The respondents were all users of the technologies and cut across several industries and business sectors. They were limited to the mainland UK. The sample size was 200, randomly selected. In each respondent firm, the officer that completed the questionnaire was either the chief technology contact or the supply chain manager of the firm.

Experienced and new researchers in supply chain product visibility, as well as practitioners, will find the content of this paper useful. It provides in one compact source a review of many of the existing hardware and software architectural frameworks for product visibility, a comparison of some of the frameworks, and the results of survey of user requirements for those systems.

4. Methods for supply chain product visibility

Supply chain product visibility is enabled by auto-identification (auto-ID) technologies. Auto-ID systems are also called automatic identification and data capture (AIDC) systems. Auto-ID systems include the barcode, RFID, biometrics (including fingerprinting, voice, facial and iris recognition, gait), optical character recognition (OCR), smart cards, and sensor technologies for location, temperature, pressure, humidity, vibration, etc. However, not all auto-ID systems are amenable to supply chain applications. Thus far only the barcode, RFID and sensor technologies for location, temperature, pressure, humidity and vibration have been used for supply chain product visibility.

4.1. Barcodes

The barcode is an optical technology for representing data. It is an optical machine-readable representation of data relating to the object to which it is attached. There are linear (or 1D) and 2D barcodes. 1D barcodes represent data by varying the widths and spacing of parallel lines. There are many types of 1D barcodes, including Code 39, Code 93, Code 128, Codabar, International Article Number (EAN), EAN-8, UPC-A, UPC-E, DataBar-14, Telepen, Interleaved 2-of-5, HIBC (Health Information Barcode), Quick Response barcodes (QR codes), PZN (Germany), IMH (Italy), MSI

(Belgium), IKS (Switzerland), coupon barcodes (EAN-13, UPC Coupon, GS1 Coupon), packaging barcodes (ITF, GS1-128, GS1 DataBar Expanded, GS1 DataMatrix), publishing barcodes (ISBN, ISSN, ISMN), postal barcodes, etc. The transformation between messages and barcodes is called a ‘symbology’. The specification of a symbology includes the encoding of the single digits/characters of the message, the encoding of start and stop markers into bars and space, the size of the ‘quiet zone’ required to be before and after the barcode, and the computation of a checksum.

2D barcodes can encode significantly more data. Among the most popular 2D barcodes are DataMatrix, PDF417, Micro PDF417, and 2D Pharma. 2D barcodes use rectangles, dots, hexagons and other geometric patterns in two dimensions to represent data.

Barcodes originally were scanned by special optical scanners called barcode readers. Nowadays, scanners and interpretive software are available on devices including desktop printers and smartphones. The barcode was first used commercially in 1966, but the very first scanning of the Universal Product Code (UPC) barcode was in June 1974. In point-of-sale (POS) management, barcode provides detailed, accurate and current information on sales, thereby accelerating decisions concerning inventory management. The two of the drawbacks of the barcode are that it is a ready-only technology – meaning that once printed, the data cannot be changed – and the scanner needs a relatively short direct line of sight to the barcode. Technologies such as radio frequency identification (RFID) have sought to replace the barcode, but the simplicity, universality and low cost of barcodes have limited the role of these other systems until the last decade.

4.2. RFID systems

RFID provides an alternative means to product identification. The main elements of an RFID system are the interrogator (or reader), the transponder (or the tag), and the middleware that collects and filters data from the reader before the data is transmitted to enterprise applications and information system. The reader and the tag do not require a direct line of sight for communication. The simplest and cheapest tags have no onboard power source. They are called ‘passive tags’; they work by relying on the very high-power energy transmitted by the reader, much like RADAR. Active tags, on the other hand, have onboard power source. They are particularly attractive in low-power RF standards such as the DASH7 (ISO 18000-7) that operates in the frequency range of 433.04–434.79 MHz (Norair, 2009). Some RFID tags have only read capability while some offer additional write memory that could serve as storage for application data such as the expiry date or date of manufacture, to facilitate data exchange where there is no network coverage.

Barcodes identify the manufacturer and product class or the stock-keeping unit (SKU), but the electronic product code (EPC) uniquely identifies each item manufactured. The EPC system is currently managed by EPCglobal, Inc., a subsidiary of GS1. The specifications for the EPC identifiers are available in the tag data standard of EPCglobal, Inc. It is an open standard. EPC is one of the industrial standards for global deployment of RFID and a central part of the EPCglobal network. The EPC network is an architectural framework of open standards developed by GS1-EPCglobal, Inc. EPC-RFID tags currently in use comply with the ISO/IEC 18000-6C standard for the RFID air interface standard.

Interest in supply chain product visibility by RFID has grown rapidly in the past decade. The technology is supported by the EPCglobal and DASH7 families of standards. Two major limitations of passive RFID systems include limited read range (on some frequencies) and detuning and attenuation effects due to materials such as

water and metals. Despite the rapidly growing academic and trade interests in RFID, the technology is unlikely to ever entirely replace both plain text and barcoding, as liabilities for the manufacturer often stipulate durable and sufficiently dependable labels. Currently, there is no one user scenario or application for which RFID has completely replaced barcoding.

4.3. Integrated RFID and sensor-based visibility systems

Most RFID systems currently in operation in supply chain applications are autonomous systems that have not been integrated with sensor technologies. But RFID systems, especially low-power RFID, can be economically integrated with other sensors (e.g., sensors for location, velocity, acceleration, temperature, humidity, pressure, vibration/shock, chemical composition, etc.). RFID needs RF silicon parts, which should ideally be as integrated as possible (e.g., using only one chip instead of two); power sources (usually batteries); a microcontroller (which usually contains a small CPU and memory); antenna for conveying RF energy; and optionally sensors (which are themselves silicon parts and hence benefit from integration). The capacity of the sensor determines the cost of the sensor and the quality of the measurements that it makes. Micro-sensors are often limited in their spatial coverage, and at a given time some of the sensor nodes might be compromised or excessive noise might affect their measurements or data channels. Some environments may also be subject to heavy signal fading.

Like RFID systems, sensor systems have the data capturing layer, a middleware that pre-processes the raw data, and an interpretation layer that transforms filtered data into information. If

the sensor has sufficient computing capacity, the three stages (i.e., data capture, middleware, and interpretation) may all be accomplished on the device itself, but often the interpretation of the data or information (i.e., decision making) is contracted to a network proxy. The way and manner RFID and sensor data are integrated are a function of the application in question. For example, the integration of RFID and sensor data is needed by concepts such as product lifecycle information management (PLIM), but not all applications require or would benefit from sensor integration.

5. Hardware and software configurations for supply chain product visibility

A visibility system is made up of hardware and software components. It usually includes a data capturing frontend, a middleware for pre-processing the data and rejecting defective reads, and a backend module for information analysis, exception handling and decision making. Visibility systems differ in many ways. Factors that may be used to distinguish them include:

- (a) The tracking technology that is used (barcode, RFID, wireless sensor networks, embedded microsystems).
- (b) The system parameters (location, temperature, humidity, vibration, etc.) that can be measured. Location can be approximated by using various technologies and techniques: GPS, eLoran, Locata, cellular networks, dead-reckoning, inertial measurements, map and/or feature mapping, WPAN, WLAN, UWB, ultrasound, infrared, short-range messaging, radio and TV broadcasts, trilateration, triangulation, received

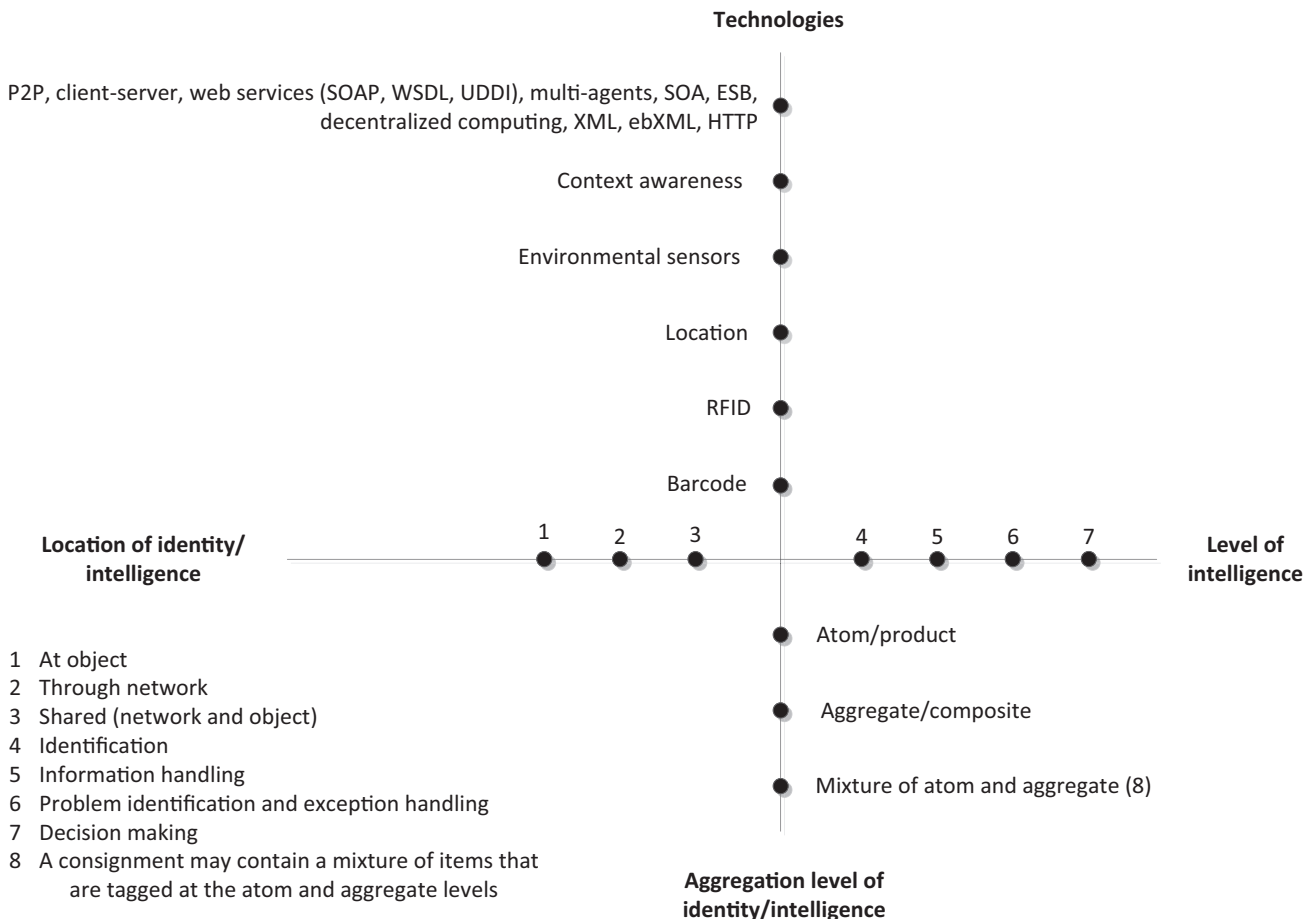


Fig. 1. A classification of visibility systems.

signal strengths, barometry (Bensky, 2008; LaMarca et al., 2004; Liu, Darabi, Banerjee, & Liu, 2007; Tesoriero, Tebar, Gallud, Lozano, & Penichet, 2010; Thiesse & Fleisch, 2008).

- (c) The locations and distribution of data capture points.
- (d) The depth of identification (e.g., at the level of the product/item/atom, composite/aggregate, lot/batch, product class, or stock-keeping unit).
- (e) Where the product information is stored (on the product itself or in a backend system; in the latter case, it is pertinent to ask how the product is linked to its surrogate in the back-end system).
- (f) The type of identification that is assigned to product (globally unique, locally unique, intra-enterprise, inter-enterprise).
- (g) The organization that is responsible for assigning the identification code.
- (h) In the case of embedded systems, the level of intelligence that is available in the embedded device. If there is not enough intelligence on the embedded device, the question arises as to whether complementary intelligence exists outside the device (Carabelea & Boissier, 2004; Cassina et al., 2008).
- (i) In the specific case of RFID systems, they differ by their frequency band, signal coding, modulation, medium access methods, supported commands, read range, data transfer rate (as determined by the reader bandwidth per channel), susceptibility to environmental interference, cost, identification speed, tag and reader form factor, tag memory, ability to

write to tag, computing and networking capacity of the reader, reader's transmission channel, whether the reader is fixed or mobile/handheld, whether the tag is passive or active, etc.

We propose the classification of visibility systems along the dimensions of technology, tagging/tracking level (atom/item or composite/aggregate), level of product intelligence, and the location of intelligence (Fig. 1).

Supply chain agility, just-in-time manufacturing and distribution, and lean operations all benefit from item-centric (or item-instance) tracking and tracing. In order to achieve efficiency through economies of scale, classical manufacturing used the lot or batch system for material control and utilization (Fig. 2). The batch operation logic, despite its shortcomings, is still the bases of some modern enterprise information systems. Such batch models do not inherently support the linking of handling data to individual material units, thereby making material-unit-level and product-level traceability an add-on feature rather than a fundamental operation logic of the production and distribution system. Lot tracing, which can be implemented in a lot system to improve visibility and control, has intrinsic drawbacks, which emanate from three main sources: physical lot integrity, data collection, and lot-process linking. The increasing complexity of manufacturing systems, customization and globalization all make it harder to maintain lot integrity: the traced lots become smaller as the variety of products increases, which in turn increases the need for lot splitting and joining. On the manufacturing shop-floor, the lot-based system of control makes it a cumbersome exercise to even mildly change product types and line configurations.

But perhaps more relevant to end-to-end visibility are the facts that the lot-based information model can result in loss of unique product information when transferring material and finished goods between supply chain partner organizations. This difficulty is caused by the use of quality classes, product mixing through lot delivery, and aggregation of products information. It does lead to rejection of good material or finished product when a part of the lot fails to meet specification. Moreover, operations can become disconnected from events, i.e., unresponsive to real-world circumstances, resulting in stock-outs, excess inventory, higher labour costs, sub-optimal asset utilization, and loss of competitive advantage. Fig. 2, adapted from Rönkkö, Kärkkäinen, and Holmström (2007), compares the batch or lot model with the item-centric model of resources and products in the manufacturing supply chain.

The means by which a product class or instance is uniquely identified and described in the supply chain constitutes its object model. See Section 7.4 and Fig. 8 for a related definition of the object model. A fundamental decision that must be made in object-modeling of items for end-to-end visibility in supply chains is the level of tagging or traceability to be adopted, i.e., deciding between item-level and aggregate-level object or entity models. It is certainly not economical to tag every item without paying attention to the economic value of the item, since the costs associated with tagging an item and ensuring end-to-end visibility may far outweigh the economic benefits of doing so. The object data and description model (Fig. 1) adopted is thus a matter of economics. A cost vs. benefit analysis of tagging is, therefore, always necessary. Where item-level model is not economically viable, the aggregate object data model should be considered as an alternative.

6. Requirements for visibility systems

There are many criteria that users need to consider when choosing visibility systems and architectures for supply chain applications. Those requirements and their relative importance are not

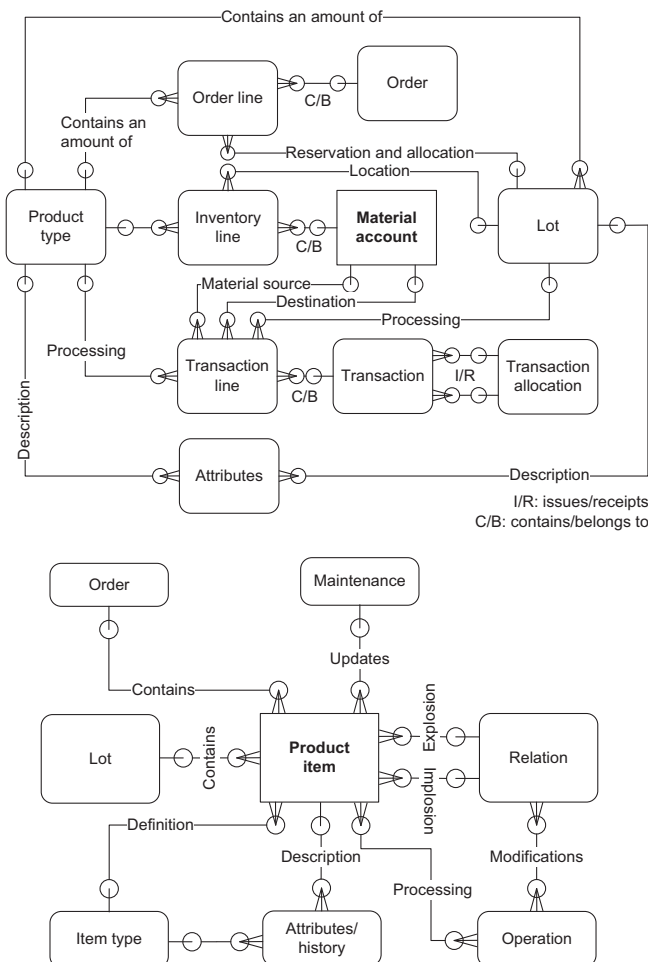


Fig. 2. Material account model vs. item-centric data model in logistics.

Table 1
Requirements for visibility systems and architectures. The average scores have the common denominator of 10 (rounded to the nearest whole number). The rows are not arranged in ascending or descending order of the scores because that will mean that the requirements with the same theme will no longer be listed in proximity of each other.

Requirement	Average score
Frequency band	6
Signal coding scheme, modulation, and medium access method	4
Supported reader commands	4
Reader read range	8
Data transfer rate (as determined by the reader bandwidth per channel)	7
Susceptibility to environmental interference (e.g., water and metals)	7
System purchase and maintenance cost	9
Tag identification speed	6
Tag and reader form factor	4
Tag memory capacity	5
Ability to write to tag	5
Computing and networking capability of the reader	5
Whether the reader is fixed or mobile/handheld	8
Provision for unique UPC (universal product code)	8
Support for hierarchical UPC. Hierarchical product codes permits efficiency in data storage and in traceability.	8
Provision of efficient (low-cost) registration mechanism for enterprise UPC	8
Cost effectiveness at the item/atomic level	9
Compatibility with EPCglobal Inc.	7
Relies on and/or extends EPC network	8
Industrial sectors for which it has been designed	8
Subscribed service (to reduce the costs of implementation and upgrades)	7
Industrial sectors where it has actually been deployed	8
Simplicity for the user community	8
Use of proprietary standards	3
Use of open standards	8
If the present system is proprietary, possibility of it evolving into an open standard in the future	6
The system and UPC are extensible on their own without external components. Extensibility allows new systems and unique product-codes to evolve	6
Support for distributed deployment within and across the supply chain	8
If your business has global reach, how do the regional/country regulations on RFID frequencies affect your business?	6
Support for intelligent objects (i.e., objects that flag exceptions and make decisions, in addition to identification)	3
Where does the intelligence lie (on the tag, in a surrogate on the network, or shared between tag and surrogate)	3
Provision of business analytics tools, i.e., inherently using events reported by the system for business decision making	9
Data mining and knowledge management must be features of the system	4
Resistant to system, tag and reader attacks	9
Resilient, i.e., failure of one node does not compromise the entire system	7
Support for privacy of enterprise's partners and customers	5
Must have special power management mechanism (for tags or embedded microsystems)	5
Must be applicable for both passive and active tags	5
Support for all major RFID reader adapters: HF and UHF readers, class 0, 0+, 1 and Gen-2	7
Support for Gen-2 features like extended data format and management of tag functionality such as memory lock, session management, password handling, tag kill, and tag erase	5
Must have a device manager capable of transparently integrating a variety of data collection technologies (barcode scanners, OCR, RFID, GPS, control actuators)	5
Ability to monitor the health and performance of deployed devices (using the device manager)	7
Must be DASH7 (ISO 18000-7) compliant. DASH7 uses low-power active RFID tags and supports tag-to-tag communication	4
Local node/enterprise should always be able to control access to centralized/network data in its databases that are accessible via EPC network	7
Provision of load balancing of communication (if network-based)	7
The inherent technologies the system uses (e.g., Microsoft .NET framework, CORBA, COM+, agents, web services, XML, ebXML, HTTP, HTML)	4
Provision of publish-subscribe messaging service between applications and readers. Because of the limited bandwidths of readers, the filtering and data collection scheme should ideally be such that reader instances only disseminate data to the message service if such data have been requested by an application	4
Ownership transfer of tags	6
Tags should always be returned to owner	5
Integration with other visibility systems via common data interfaces	4
The system must accept heterogeneous data standards, syntaxes, semantics and databases of trading partners	7
It should be able to track and trace processes and activities, in addition to lot and product tracking	5
Suitability for closed-loop product lifecycle information management (PLIM)	5
System's future prospects in terms of market needs and cost	8

available in any single source. In order to develop criteria for assessing the various visibility systems we conducted a survey by questionnaire of supply chain managers in the UK to determine their lists of priorities of the features of a supply chain visibility system. Since the aim was to gauge user needs, the survey was limited to users of the technology; vendors were not sampled. In each respondent organization, the person that answered the questionnaire was either the chief technology/RFID contact or the supply chain manager of the establishment. The result of the survey is shown in Table 1. Another aim of the survey was to determine the criteria to use in comparing the visibility systems. The systems

described in Sections 7.2–7.5 have been compared using some of the criteria of Table 1. The outcome has highlighted some important and interesting issues from the perspective of users of the technologies. In particular, it indicates the considerations that matter most to the users. It is clear that the majority of the users want a relatively simple system that answers their present needs. Whilst issues like the reader range, acquisition and maintenance costs, whether the reader is stationary or mobile, resistance to attacks, the pedigree of the system, and the possibility of distributed deployment across the firm and the supply chain are important today, matters like tag and reader size, tag memory capacity, support

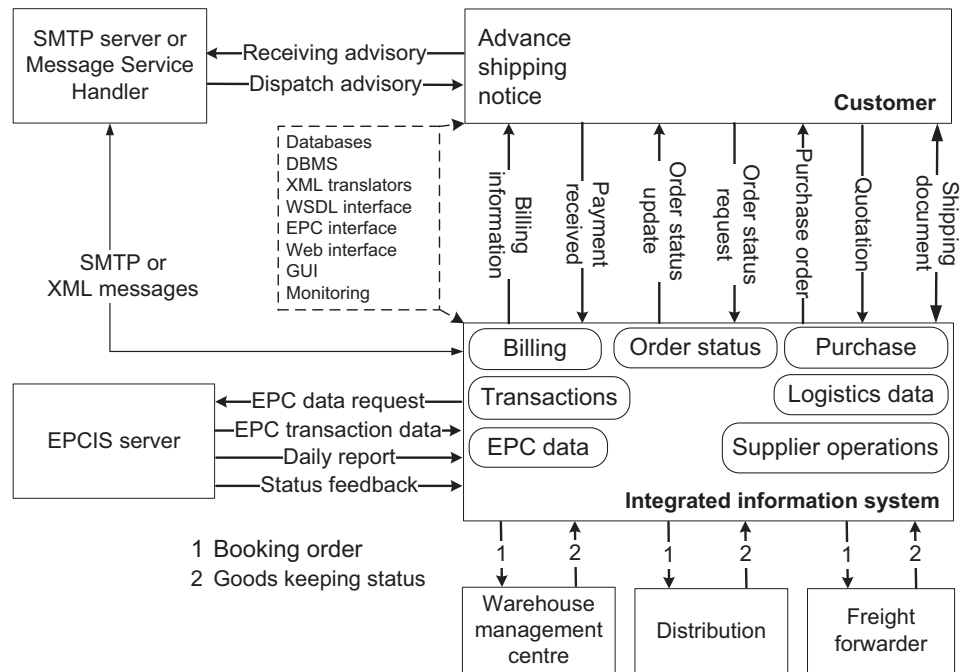


Fig. 3. Information flow across visibility-enabled supply chain.

Table 2

References for visibility systems and architectures.

Examples of commercial systems

Proprietary mobile device is part of the system
No proprietary mobile device is part of the system

Examples of academic systems

A new mobile device is part of the system (see Section 7.1)

No new mobile device is part of the system

Agaidi (2010), Omnitrol Networks (2010), PearTrack (2013), and Savi Networks (2013)
EPCglobal Inc. (2004a, 2004b, 2006a, 2006b, 2010a, 2010b), Microsoft. (2013), OATSystems (2006),
Odin (2013) and Sun Microsystems Inc. (2005)

Hsu and Liao (2011), Hsu and Wallace (2007), Kwok et al. (2010), Martínez-Sala et al. (2009),
Mattoli et al. (2010), Oliveira et al. (2012), Papetti et al. (2012), and Wang et al. (2013)
Amaral et al. (2011), Azuara, Tornos, and Salazar (2012), Bardaki, Kourouthanassis, and Pramatri
(2011), Bechini et al. (2008), Botts (2007), Cassina et al. (2008), Chatziantoniou et al. (2011), Chen,
Chen, Chen, and Chang (2007), Chen, Hsu, Liu, and Yang (2012), Chen, Zhu, Hu, and Ku (2011),
Cimino and Marcelloni (2011), Floerkemeier et al. (2007), Främling et al. (2006), Gomez et al.
(2012), Hong, Kim, and Kim (2010), Hribernik, Rabe, Thoben, and Schumacher (2006), Hsu et al.
(2007), Huang and Chang (2011), Jakkhupan, Arch-int, and Li (2011), Kärkkäinen and Holmström
(2002), Kärkkäinen et al. (2003, 2004), Kiritsis (2011), Ko et al. (2011), Oztekin, Pajouh, Delen, and
Swim (2010), Prabhu, Su, Ramamurthy, Chu, and Gadh (2005), PROMISE (2008), Römer, Schoch,
Mattern, and Dübendorfer (2004), Shin, Chin, Yoon, and Kwon (2011), Spieß, Bornhövd, Lin, Haller,
and Schaper (2007), Sundaram, Zhou, Piramuthu, and Pienaar (2010), Thakur and Hurburgh (2009),
Véronneau and Roy (2009), Vrba et al. (2008), Woo, Choi, Kwak, and Kim (2009), and Xu et al.
(2011)

for intelligent products, knowledge management, and the system's underlying technologies are less relevant.

Concepts like closed-loop product lifecycle information management (PLIM) are yet to be considered a priority by the practitioner community despite a lot that has been pursued in that direction by academics (Cassina et al., 2008; Meyer, Främling, & Holmström, 2009; Kiritsis, 2011; Ranasinghe et al., 2011) and their future potential. On other hand, there are services that are not yet available (such as the ability to track and trace processes and activities, in addition to lots and products) that the practitioners consider presently useful, if not essential.

7. RFID and sensor-based visibility architectures

There are many open and proprietary schemes and products for achieving end-to-end visibility in supply chains. In Sections 7.2–7.5 we review some of the best known models and products.

Fig. 3 is meant to depict the representative nature of enhanced information flow across the supply chain resulting from visibility-enabled supply chain. Table 2 captures the main characteristics of the configurations and structures discovered from an extensive survey of academic and trade literature. Some of the systems in Table 2 simply provide middlewares instead of full deployment architecture. Only those architectures that have actually been piloted are cited in Table 2. In order to reduce space, we have cited only one reference for each system or middleware.

Since the architectures are many, only four of them are described in some detail below (Sections 7.2–7.5). The four have been chosen based on the fact that they are applicable across industrial sectors and have enjoyed the widest industrial adoption.

The majority of the academic systems have not been given names by their originator references. Hence no names have been provided for them in Table 2. Some of the systems and architectures have been developed for specific industries but they are

extensible to other industries. For example, the system of Papetti et al. (2012) was proposed for the food industry but it can be adapted to other industrial sectors. Systems by PearTrack (2013) and Savi Networks (2013) come with special tags for applications in freighting and international shipping.

7.1. Academic systems with new mobile devices

The commentary in this section pertains to academic architectural frameworks for supply chain product visibility (Table 2).

Hsu and Liao (2011) chose CF-1700T reader by Sunion, which has an operating frequency of 13.56 MHz, as the thin mobile client. It is an RFID reader with a CF (Compact Flash) transmission interface. The RFID reader is attached to the PDA with a CF slot. The power of the reader is supplied by the PDA with a consumption rate at 15 mA (max) in operation and 4 mA in sleep mode. RFID tags complying with ISO15693 standard can be detected by the reader. The reading/detection distance is up to 4 inches. The PDA used by Hsu and Liao (2011) is of ASUS A716 Series with an Intel Xscale PXA255 400 MHz processor. It runs on a Microsoft Windows Powered Pocket PC 2003. The PDA has a CF Type II slot to plug in the RFID reader and can access wireless LAN through WiFi.

The architecture of Hsu and Wallace (2007) was field tested by equipping each test vehicle with a real-time guidance package (RT-RGP) and a pocket PC operated together with a 3G wireless card and a Bluetooth-enabled GPS device. The co-pilot and the RT-RGP operated on the pocket PC. The GPS device was used to locate the vehicle and generate guidance data. Bluetooth was used to connect the pocket PC to the GPS device, and the co-pilot's voice communication guided the vehicle driver. The co-pilot software managed the data transmission and provided the route guidance. The 3G card connected the probe system to the Internet and transferred data between the server and the various onboard units.

The scheme by Kärkkäinen, Holmström, Främling, and Artto (2003) and Kärkkäinen, Ala-Risku, and Främling (2004) does not require a special/new mobile device. It revolves around an application that was built on two software components. The first is a client agent, which is installed at each "checkpoint" where shipments are tracked. The second component is a server agent that resides at the computer specified by the URI-part of the ID@URI identity scheme (Främling et al., 2006). It receives tracking messages from the client agents and forwards the tracking information to the in-house tracking system of the participating organization. When a consignment arrives at a checkpoint, its tracking code is read to the client component. The client then communicates the identity of the checkpoint, the tracking code of the shipment and the time of the reading operation to the server. The server compares the tracking code to the consignment's identity used in the in-house tracking system of the organization, after which the server updates the information in the tracking system of the organization.

In the scheme of Kwok, Ting, Tsang, Lee, and Cheung (2010), passive Class-1 Gen-2 RFID tag and RFID reader embedded PDA were used to develop a mobile RFID environment. Furthermore, a PC was employed as the system server. It allowed high speed access and provided large storage capacity. Hence it could accommodate the need for many accesses as well as manipulate large databases. The system could be accessed remotely via WLAN. An RFID label printer was used to print the EPC tag of each product. The software packages used to implement the system included a database management system (DBMS) for data manipulation, and a Microsoft .NET framework for developing the applications.

Martínez-Sala, Egea-López, García-Sánchez, and García-Haro (2009) developed a visibility and information management scheme for a packaging, transport and display system called the MT. The MT is designed for the grocery supply chain. It serves as packaging at

the producer, as a transport unit for product distribution, as storage at warehouses, and as a display stand at the retail outlet, all in the same mechanical system. It is thus a returnable packaging and transport unit. The system by Martínez-Sala et al. (2009) relies on active RFID tags. Martínez-Sala et al. (2009) claimed that commercial active RFID systems are expensive and lack the flexibility needed to customize MT and develop future services. Therefore, they designed and developed an active tag from commercial off-the-shelf components. They used Texas Instrument CC2510F32 System-on-Chip (SoC) as the active tag platform, operating at 2.4 GHz. The functionality implemented on the platform is programmable temperature logging and generic data memory reading/writing. In addition, since battery depletion is a major concern for active tags, they developed and implemented an energy-conserving anti-collision procedure. They plan to eventually have a family of readers with different capabilities. Their present readers are composed of two components: an embedded computer and an independent RF interface. This design follows that of WiFi access points: the independent RF module performs communication with MT tags, whereas the embedded computer implements functionality such as data filtering and aggregation and communication with the middleware and reader hubs. The RF module identifies and interrogates tags; it also sends data to the embedded computer via a serial interface (either RS232 or USB).

The hardware of Mattoli, Mazzolai, Mondini, Zampolli, and Darío (2010) is a tag-datalogger for monitoring the physical states of food and goods during transport, storage and vending. The miniaturized device integrates temperature, humidity and light sensors and a microcontroller. The microcontroller manages the sensors in a power-safe mode and records the ranges of the measures on its memory. The transmission of data is based on an infrared communication, making it possible for the device to communicate with the most common personal devices, such as Smartphones and PDAs with integrated infrared port.

The system of Oliveira, Noguez, Costa, Barbosa, and Prado (2012) was aimed at determining vehicle positions for various logistics applications, including route planning, monitoring the path of the vehicle/cargo, and route adherence enforcement. It uses low cost off-the-shelf mobile clients. It experimented with the clients Motorola Dext Motoblur, HTC Desire, and ZTE V821 and opted for Motorola Dext Motoblur because of its superior performance in terms of location errors.

The work by Papetti et al. (2012) sought to associate information on cheese quality parameters with the RFID traceability. It developed and tested a system that integrates an RFID tracing system with a non-destructive quality analysis system for each single unit of cheese product. This system records the two types of information on a web platform. This was carried out in order to develop a scheme of quality tracing and information dissemination. Two types of RFID tags were used in the experiments: TAG cheese HF2009 and TAG cheese arrow screw. The reader used was CPR.MR50-USB Multi ISO (ISO14443-A/B, ISO15693 & NFC).

The SPIDER (Supply chain Pedigree Interactive Dynamic Explore) system of Wang, Ting, and Ip (2013) has an RFID-based cargo monitoring system, which was used to collect real-time data about the physical states of the cargo for transmission to SPIDER. SPIDER is a platform for the public and governments to explore the traceability information at each stage of the food supply chain and provide an easy implementation of Hazard Analysis and Critical Control Point (HACCP) in the food industry. SPIDER is said to be a secure and global platform to verify, inspect and, if necessary, investigate food history and pedigree effectively and efficiently. HACCP, as opposed to finished product inspection, is a standard for food product safety that aims to prevent food contamination along the supply chain.

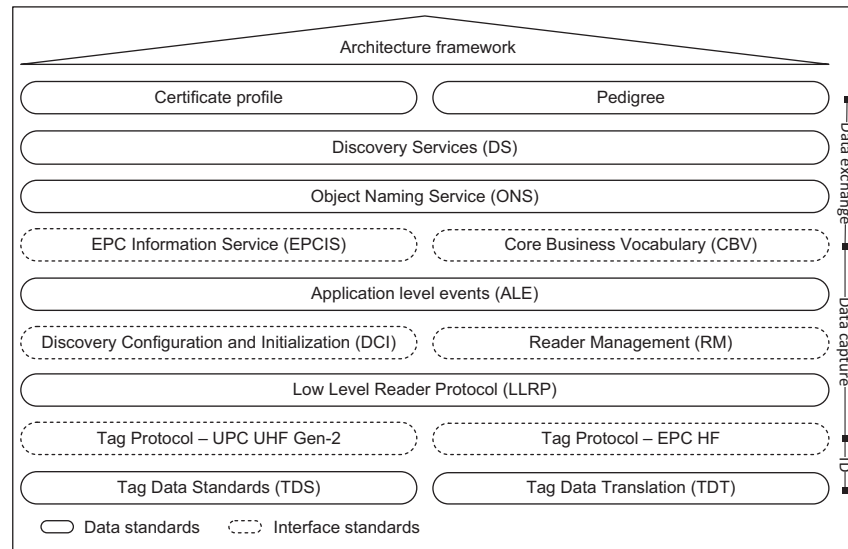


Fig. 4. EPCglobal architecture framework.

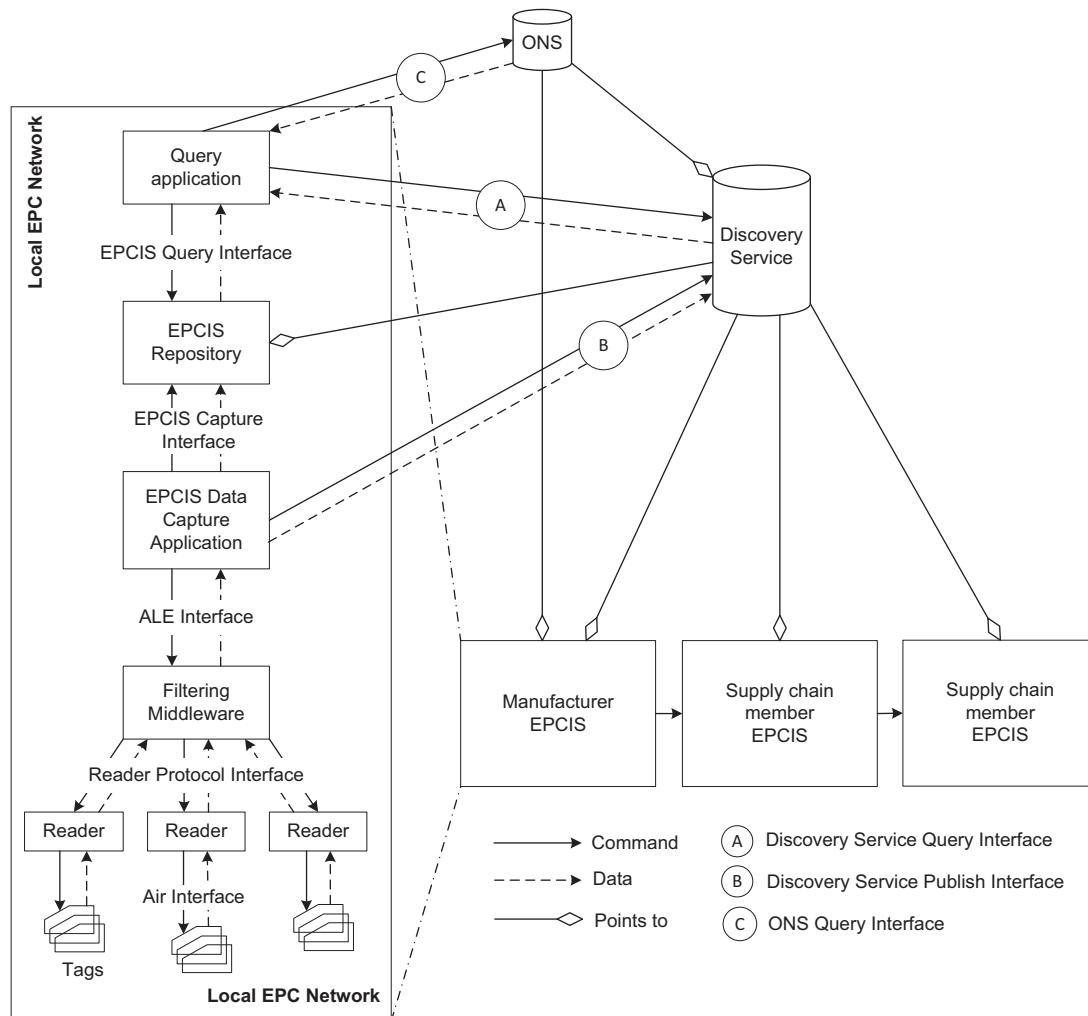


Fig. 5. A representation of the standard interfaces of EPC network.

7.2. The EPC network

The EPC network was developed by EPCglobal Inc., a joint venture between GS1 (formerly known as EAN International) and GS1

US (formerly the Uniform Code Council, Inc.). EPCglobal Inc. is an open, worldwide, not-for-profit, consortium of supply chain partners working to encourage and direct the global adoption of the EPC network. The EPC network is a neutral entity that leverages

RFID and the Internet to collect and communicate real-time data and create visibility about individual products as they transit supply chains. It provides a lineage of products movement that can be accessed by authorized parties. It uses firewalls, encoding, and other safe measures to guarantee the security of the transmitted information. The EPC network also provides RFID-related standards for data generation and transmission. The architecture framework for EPCglobal's standards is shown in Fig. 4 and the EPC network interfaces are depicted in Fig. 5 (EPCglobal, 2004a, 2004b, 2006a, 2006b, 2010a, 2010b).

The EPC network securely connects servers containing information related to products identified by the electronic product numbers (EPC). The servers are called EPC Information Services (EPCIS) and are linked via a set of network services. Each participant in the EPC network stores relevant information relating to specific electronic product code (EPC) numbers in their own EPCIS servers. In some circumstances, local databases provide the information that is needed by the EPC network member. In situations where local databases cannot provide the required information, inquiries trigger searches in the electronic registries of business partners via the EPC network. The EPC network provides a global network model and infrastructure support for identifying, locating and managing inventory in the global multi-supply chain by RFID (EPCglobal, 2004a, 2004b; Shih, Sun, & Lin, 2005). In a typical application, palletized items of merchandise are tagged with RFID transponders whose details (product identities) are read by the interrogators located within the premises of the manufacturer enterprise. The records are then securely transmitted via the Internet to the EPC network and to the receiver enterprise. As soon as the vehicle loading is accomplished, to complete the bill of lading (BoL) the cargo in the vehicles may be scanned by the onboard RFID readers. The data are then shared, by secure transmission through the Internet, between the logistics service provider (the carrier), the shipper, the receiver, and the EPC network.

In the model of Fig. 5, a mobile worker may be added. The premise for that is that while out in the field, the mobile worker can use an SMTP-enabled personal digital assistant (PDA) to access information from EPCglobal and from enterprise databases. When there is no connection, the mobile worker can still perform critical tasks using the limited database and computing capacity on the PDA, and later (when there is a connection) transfer the information onto enterprise servers.

7.3. The SaviTrak system

The SaviTrak system is owned and operated by Savi Networks (www.savinetworks.com). For the purpose of illustration we abstract the SaviTrak system in Fig. 6. Our discussion of the SaviTrak

network is based on this abstraction. The system provides a network infrastructure for sharing location information within and across supply chains. In Fig. 6, location-enabled products act as information carriers between supply chain partners. Software components located at different places in the supply network can communicate using web services.

SaviTrak combines GPS, RFID, and sensor-based information network infrastructure and is available in all major ports and terminals of the world. Outside the geographies covered by the SaviTrak network of RFID and sensor checkpoints, assisted-GPS is used to locate cargo, and cellular or satellite technology is used for communications. The SaviTrak system monitors and issues alerts concerning tampering or forced-entry of containers, shipments moving into incorrect zones or being improperly located, shipment's physical states (temperature, humidity, light or shock) exceeding environmental thresholds, shipments deviating from intended workflows and other user-defined exceptions. The operational environments involve customs authorities around the world, major international shippers, terminal operators, and third-party logistics service providers. Savi Networks claims that, to date, the SaviTrak network covers most of the major ocean transportation ports and terminals and rail heads in Europe, North America, Australia, China, and along the Mediterranean coast. SaviTrak boasts of over 4000 sensor checkpoints in more than 50 countries. For in-transit visibility between checkpoints, SaviTrak accepts feeds from onboard RFID readers, GPS units and mobile satellite communication modems attached to a power source such as a truck cab, ocean vessel or rail engine.

As the goods make their way to the receiver from the shipper (see Fig. 6), the status of the cargo can be monitored onboard the vehicles and the information transmitted in near real-time to all the cargo's stakeholders. Since the vehicles are continually tracked throughout their journeys by the carrier (or a contractor) using assisted-GPS, the locations of the in-transit goods can always be inferred from those of the vehicles irrespective of the stage of the journey. During transshipment, at all major sea and land ports that have installed RFID reader infrastructure, the tags on the goods can be scanned and the results transmitted to all the partners with stake in the cargo and to EPCglobal databases. At sensor checkpoints, the locations of the goods are inferred from those of the RFID readers.

SaviTrak claims to use a robust tag device and cellular roaming agreements in over 150 countries to provide global visibility and mobile connectivity to in-transit inventories of its customers. The multi-wireless tag device, which clamps onto a shipping container door, receives assisted-GPS location information at user-configurable time intervals. Sensors embedded in the tag detect and record security breaches of the container doors as well as changes of temperature, light and humidity inside the container. The device then communicates the information to either active RFID, cellular or satellite communications networks using GPRS.

7.4. Microsoft BizTalk RFID

Microsoft BizTalk Server is an Enterprise Service Bus (ESB). The Microsoft BizTalk RFID is built on the Microsoft .NET architecture and has both Server RFID and Mobile RFID. Microsoft BizTalk Server RFID enables users to build and deploy RFID solutions by: decoupling of design and deployment activities; agile and flexible process architecture with plug-in components for event handlers and workflow at the frontline of the enterprise; managed application programming interfaces (APIs) and tools for process lifecycle management; scalable event handling infrastructure with hooks to standards that support RFID and sensor technology; and legacy integration with BizTalk Server for end-to-end solutions for all enterprise and supply chain processes.

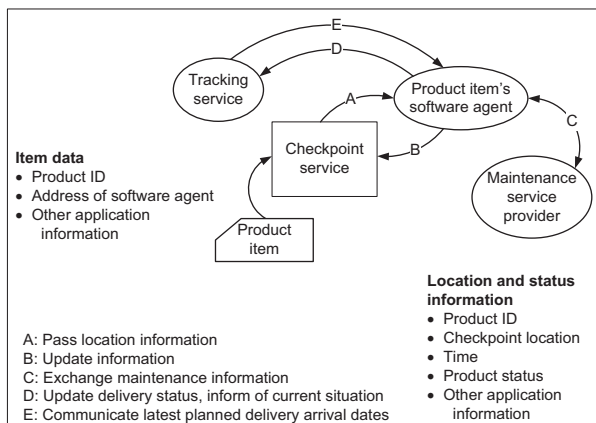


Fig. 6. An abstraction of SaviTrak.

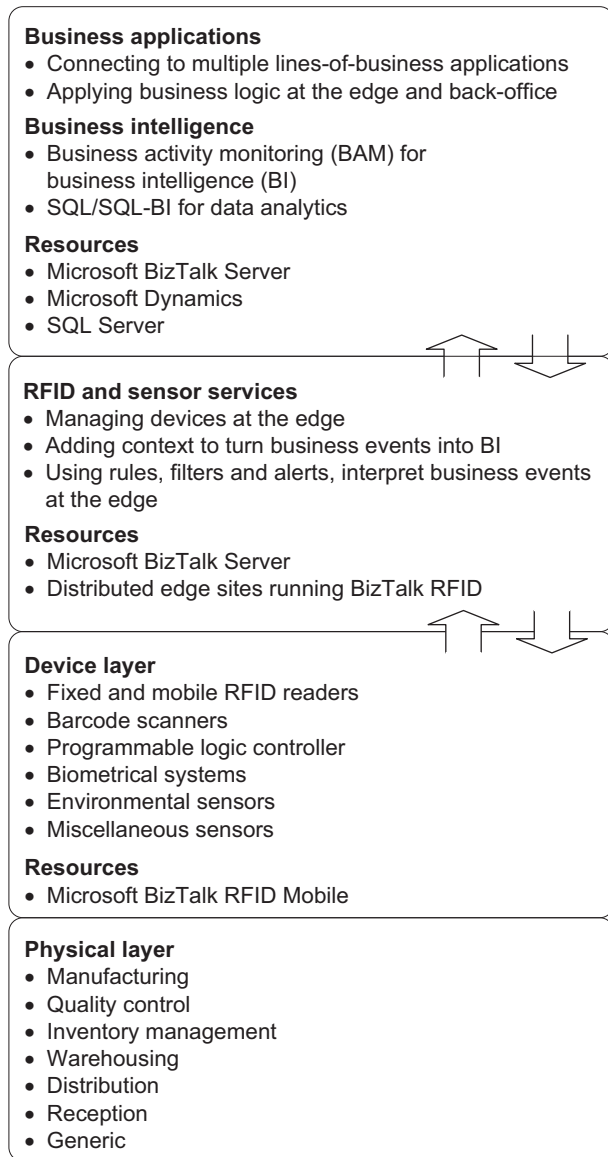


Fig. 7. A schematic of Microsoft BizTalk RFID logical stack.

Microsoft BizTalk RFID Mobile is an extension of the Microsoft BizTalk Server RFID platform to thin mobile clients. BizTalk RFID Mobile is an extensible platform for developing, deploying and managing sensor applications (RFID, barcode, WiFi) on mobile devices.

Figs 7 and 8 show the logical layers of Microsoft BizTalk. The device service provider interface layer is made up of an extensible, general-purpose application programming interfaces (APIs) that enable software developers to create specialized interfaces that permit RFID devices to function harmoniously in the Windows OS environment. In order to allow for effortless integration of “software devices” (i.e., the specialized interfaces that permit RFID devices to function seamlessly), Microsoft provides the platform, specifications and test packages by way of an RFID software development kit (SDK).

The engine and runtime layer uses event handlers and business logic to enable applications to filter, aggregate and map raw sensor data into business information and intelligence. The event processing engine is agnostic of sensor type and the sensor's communication protocol. The engine allows application developers to create and manage logical sensor processes and events indepen-

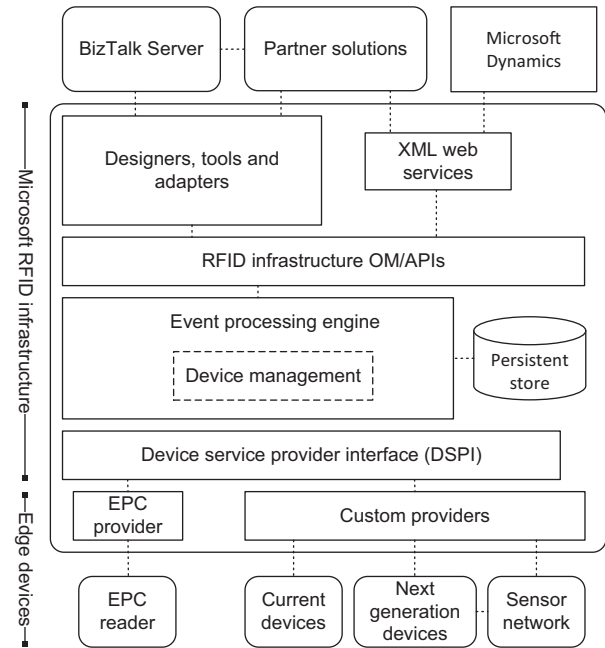


Fig. 8. The inside of Microsoft BizTalk.

dent of the underlying peripheral device type. The engine defines command sets for specific peripheral device types. The engine can interface with almost any device.

The central part of the processing engine is the “event processing pipeline”. The event execution pipeline processes events either synchronously or asynchronously. It provides a means to process and execute RFID events by grouping readers into logical associations. The RFID Object Model (OM) provides a mechanism for creating event processing pipelines or trees of extremely simple or complex profiles. The event processing engine also contains the event handler, an extensible component that allows distributed RFID events to be processed on the chosen business logic. The engine and runtime layer also contain the device management block.

Another layer of Microsoft BizTalk RFID is the OM/API layer. The OM and APIs are meant to assist developers in designing, deploying and managing RFID and sensor solutions. The Object Model includes the device management module, process design and deployment, event tracking and health check. It provides the APIs to rapidly construct and deploy RFID processes.

The Microsoft BizTalk RFID server also has a “designers, tools, and adapters” layer. Designers consist of an array of tools to help developers compose different strands of business processes, while adapters assist in integrating real-time RFID events with Microsoft BizTalk RFID server and line-of-business applications.

7.5. Sun Java System RFID

The Sun Java System RFID software architecture is built with Java Enterprise System software and technology, which provides an integrated set of enterprise infrastructure services (Sun Microsystems Inc., 2005). It is marketed as an enabler of EPC network. The architecture stack (Fig. 9) has RFID readers and sensors at the bottom. Reader and sensor reads are fed into the next layer of the stack (Java System RFID Software) for processing. The Java System RFID Software has two main parts: the Java System RFID Event Manager and the Java System RFID Information Server. The Event Manager filters and aggregates streams of tag or sensor reads before sending the data to a requesting application. The Event Manager follows the EPCglobal standards

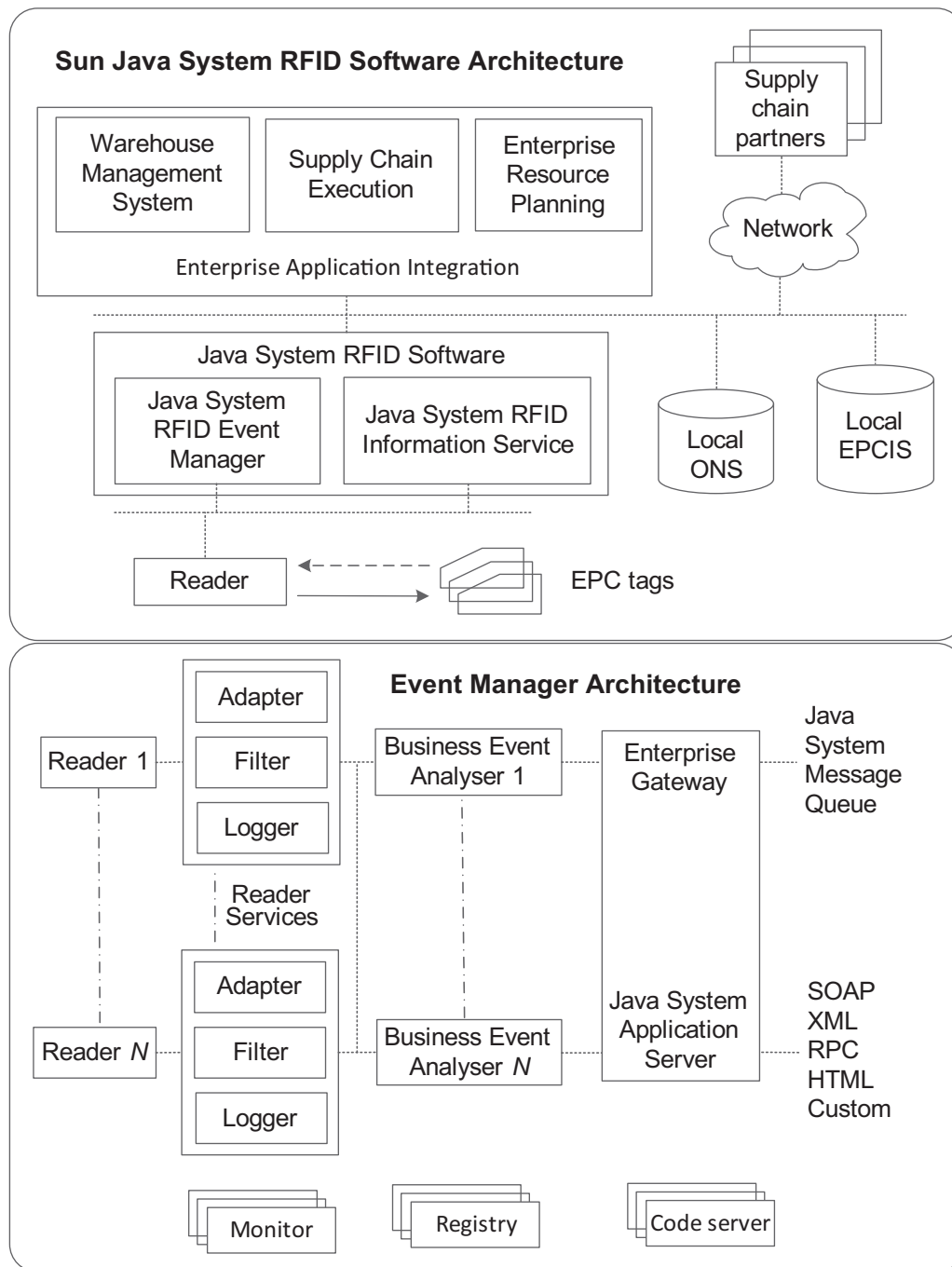


Fig. 9. Java System RFID architecture.

and provides additional features and utilities needed by large-scale enterprise applications.

The Java System RFID Information Server allows applications to access business events produced by the Event Manager. It also is the integration layer that provides several preferences for integrating the Event Manager with enterprise information system – enterprise resource planning (ERP), warehouse management system (WMS), and supply chain management (SCM) – and other enterprise applications that need the EPC data and information. Data from the Event Manager is fed into the Information Server, where it is warehoused, reformatted and/or sent to any application that needs it.

The Sun Java System RFID software architecture has support for Web Services Description Language (WSDL), Universal

Description, Discovery, and Integration (UDDI), and SOAP. In Fig. 9, the device adapter allows devices from different manufacturers to communicate and interact with the RFID Event Manager. Loggers are similar to device adapters but are deployed to notify external systems of RFID and non-RFID events. The Enterprise Gateway serves as the common interface for enterprise application demanding data from the RFID Event Manager.

8. Comparison of product visibility architectural frameworks

In order to compare the RFID deployment architectures of Sections 7.2–7.5 we used some of the criteria of Table 1 and determined the extent to which the designs satisfy those needs.

Table 3

Comparison of RFID deployment architectures.

Requirement	EPC network	Microsoft BizTalk	Sun Java System	SaviTrak
Is the system an open or proprietary standard?	Open	Proprietary	Proprietary	Proprietary
If the present system is proprietary, what is the possibility of it evolving into an open standard in the future?	Open system	No prospect of evolution into an open system	Same as for Microsoft BizTalk	Same as for Microsoft BizTalk
Subscription charges?	Yes	No, except (i) for software upgrades or when software is provided as a service (SaaS), and (ii) for registration and subscription to EPC network	Same as for Microsoft BizTalk	Yes
Comes with hardware and software?	No hardware is supplied to users apart from the network	Yes	Built on Java Enterprise System software and technology	Yes, comes with tracking tags and connection to Savi network
Compatible with EPCglobal (possibility of connecting to EPC network)?	Inherently EPC	Yes, can link to EPC network	Yes, can link to EPC network	Yes
Accepts sensor feeds other than RFID?	No	Yes	Yes	Yes, but network-dependent. GPS is advertised as an available data type
Industrial sectors for which the system has been designed	Generic	Generic	Generic	Mostly logistics and transportation, especially international shipping
Industrial sectors where the system has actually been deployed	Generic	No information	No information	Logistics and international shipping
Provision for globally unique UPC (universal product code)	Yes. Centrally allocated firm-level ID; product-level ID decided and controlled by the firm	Yes, based on EPC	Yes, based on EPC	Yes, based on EPC
Support for custom ID and hierarchical UPC	Uses EPC. The identifier includes manufacturer, product class/type, and the product instance itself. Internal ONS could be used to support custom EPC, but no headers yet exist for custom identifiers. Version 1.4 of EPC supports several coding schemas (see below)	Supports a variety of coding schemes such as all EPC formats and custom/private	Same as for Microsoft BizTalk	Uses EPC and its own coding scheme
Provision of efficient registration mechanism for enterprise UPC	By subscription to EPCglobal	Both intra-firm custom registration and subscription to EPCglobal are possible	Same as for Microsoft BizTalk	Same as for Microsoft BizTalk
Cost effectiveness at the item/atomic level	Small serial numbers support cheap tags, but non-RFID data sources are not supported	Support cheap EPC tags and non-RFID data sources	Same as for Microsoft BizTalk	Same as for Microsoft BizTalk
Simplicity for user community	Simplicity is at the root of EPC network, and the network address is part of the header of EPC (see below)	Permits the combination of EPC with other coding schemes like the DIALOG (ID@URI) or custom policies	Same as for Microsoft BizTalk	Basically EPC
Support for distributed service and name resolution	Not yet distributed: only one root ONS exists	Distributed, but information lookup is still subject to node availability	Same as for Microsoft BizTalk	Distributed but lookup is available only within Savi Networks
Resilience, i.e., failure of one node should not compromise the entire system	Failure at one node can affect lookup to that node but not to data already in the EPC network	Resilient to changes in network topology. Event Manager implements federated service architecture	Same as for Microsoft BizTalk	No information is available
Reader fault and configuration management	No	Yes	Yes	No information is available
Support for intelligent objects (i.e., objects that flag exceptions and make decisions, in addition to identification)	None	The software architecture can easily support object intelligence, but no embedded devices with intelligence are currently offered	Same as for Microsoft BizTalk	EPC, GPS, and environmental and vibration sensors are part of the array, but there is currently no onboard or surrogate intelligence
In-transit limitation	Visibility is unavailable between checkpoints except if a reader with a communications link is onboard the vehicle	Same as for EPC network	Same as for EPC network	Same as for EPC network
Class-product-instance or containment relation	EPC identifies a class of objects or a product instance. It cannot ordinarily link a product instance to its class unless it is possible to write this information to the tag	Same as for EPC network unless a special tag or an embedded microsystem is used	Same as for Microsoft BizTalk	Same as for EPC network unless a special tag or an embedded microsystem is used
Provision of direct business analytics tools, i.e., inherently using events reported by the system for decision making	No, although EPC network has a business vocabulary standard (Fig. 4)	Yes	Yes	Basically a track, trace and alert system

(continued on next page)

Table 3 (continued)

Requirement	EPC network	Microsoft BizTalk	Sun Java System	SaviTrak
RFID data-event mining and knowledge management is a feature of the system	No	Not stated	Not stated	No
Extensibility of system for new data types and business analytics	No	Yes	Yes	Not clear
Support for privacy of end-users of tagged products	The air interface protocol standard contains the tag kill command	Not stated, but may be inherited from EPC network	Same as for Microsoft BizTalk	Same as for Microsoft BizTalk
Has special power management mechanism (for tags or embedded microsystems)	The issue is not part of the EPC network	Sensor feeds are admissible but device power management is not part of the system	Same as for Microsoft BizTalk	No information is available
Is applicable for both passive and active tags	Yes	Yes	Yes	Yes, the tags are by Savi Networks
Local node/enterprise can always be able to control access to centralized/network data that it has contributed	Yes, via the local ONS and EPCIS	Same as for Microsoft BizTalk	Same as for Microsoft BizTalk	Same as for Microsoft BizTalk
Provision of load balancing of communication (if network based)	None	Not stated	Not stated.	Not stated
Support technologies the system uses (e.g., CORBA, COM+, agents, web services, XML, ebXML, HTTP, HTML)	XML, HTTP, HTML	COM+, XML, ebXML, HTTP, SOAP	JMS queue, XML, ebXML, HTTP, SOAP	Not stated
External sensors can be used to trigger readers.	No	Yes	Yes	Possibly
Provision of publish-subscribe messaging service between applications and readers	No	No, but the event manager can readily be upgraded to support the feature.	Same as for Microsoft BizTalk	The details of the event manager are not in the public domain
Provision of ownership transfer of tags	No	No, but can be added through software support	Same as for Microsoft BizTalk	The inside of SaviTrak is not in the public domain
Integration with other visibility systems via common data interfaces	Yes, through one of the data interfaces	Yes	Yes	No public information is available
The system accepts heterogeneous data standards, syntaxes, semantics and databases of trading partners	No	Yes	Yes	No
Able to track and trace processes, activities and actors, in addition to lot and product tracking	No	No	No	No
Provision for closed-loop product lifecycle information management (PLIM)	This requires some level of onboard or delegated intelligence, which does not yet exist at a cost that will encourage supply chain-wide adoption	Same as for EPCglobal	Same as for EPCglobal	Same as for EPCglobal

Table 3 contains the comparisons. In that table, only a few of the comparisons concern identifier structure, since Microsoft and Sun Java platform are compatible with many RFID standards, EPCs, and sensor arrays. It is important to state that the essence of this section is not to make value judgment of the architectures but to discuss the properties of the existing and proposed deployment networks and designs. Hierarchical product codes permits efficiency in data storage and in traceability. EPC network appears to be progressively adopted by the industry and the other systems in Table 3 all support and extend EPC network.

Microsoft BizTalk RFID, Sun Java System RFID and SaviTrak are all extensions of EPC network, but while the extensibility of BizTalk RFID and Sun Java System to sensors beyond RFID is well described in the existing literature, the design and applications of SaviTrak seem to have been directed at one specific industry, namely, logistics, especially international shipping.

Most of the entries in Table 3 are self-explanatory. Anyway, the full explanation of every row of the table requires another paper on its own. Hence, the following paragraphs provide annotations on only some of the contents of the table.

In the systems compared in Table 3, only the EPC network uses open, worldwide standards that are supported by GS1; the other systems are understandably (for commercial reasons) proprietary.

EPC standards are developed by the community of technology providers and end users. Open systems permit wider market acceptance and technology growth via a larger developer community. Also, EPC tag codes are unlikely to change in the lifecycle of a product: the lookup method EPC network uses makes changes to codes improbable. However, since the other architectural frameworks (Microsoft BizTalk RFID, Sun Java RFID, and SaviTrak) are linked to and are compatible with EPC network, they stand to benefit from any advances in the technology and outreach of EPC network. On the other hand, initial registration and subscription to EPC network may raise the entry cost for some enterprises, especially small and medium sized enterprises (SMEs).

EPC network develops and publishes EPC standards (see Fig. 4). It also provides a registration and subscription service, but it does not offer hardware to its subscribers. For a given EPC, the EPC network uses the object name service (ONS) as a lookup scheme to locate URLs where information could be found. ONS uses naming authority pointer (NAPTR) domain name service (DNS) files. ONS is a scalable hierarchical lookup mechanism that uses existing DNS tools and protocols to match information. ONS entries provide URLs of information services as well as the meta-data of the information (web pages, EPC information services, web services, XML data files, etc.). ONS determines the Manager ID, which usually

points to the manufacturer of the product. The first-tier ONS is provisioned by EPCglobal, Inc. The second-tier ONS is implemented within a company by using in-house DNS name server. It provides for the location and resolution of different product classes or SKUs within the company. Records in the root-level ONS lookup system are provided to subscribers of EPCglobal, Inc.

The EPC network's standards (Fig. 4) are essentially RFID-centric, in which case the device layer is rather simple: it is a set of extensible, generic application programming interfaces (APIs) that allow different RFID readers to work together seamlessly. EPC tags attached to objects are read and optionally written to by readers. Active tags are also able to provide additional sensor data such as the current value or history of a parameter. However, for both passive and active tags the communication is always initiated by the reader. Moreover, the tags do not communicate directly with each other – unless they are DASH7 capable. In the case of Microsoft BizTalk RFID and Sun Java RFID platforms, the device layer is able to discover, configure and manage any device on the network, including optical scanners, RFID readers, printers, sensors, and smart devices that host multiple sensors.

The SaviTrak system has been designed for logistics and transportation, especially international shipping. The other systems are more versatile, i.e., they can be deployed in any industry where there is a need for product visibility. However, SaviTrak provides a dedicated global and networked infrastructure (with checkpoints at shipping ports) that is different from that of the EPC network and that both Microsoft BizTalk RFID and Sun Java RFID do not currently offer.

The EPC has been designed to meet the needs of various industries as well as assure uniqueness for all EPC-conforming RFID tags. Some of the existing GS1 identification types (such as the Global Returnable Asset Identifier – GRAI) already provide for unique identification of individual objects. However, the Global Trade Item Number (GTIN) only identifies the product class or SKU rather than an individual instance of a particular product class. To ensure that an EPC always uniquely identifies an individual product in the case of a GTIN, the EPC is created as a Serialised Global Trade Item Number (SGTIN) by combining a GTIN product identifier with a unique serial number.

The official representation of an EPC is a URI (uniform resource identifier). URIs can be classified as uniform resource locators (URLs), as uniform resource names (URNs), or as both. In EPC, URI is the 'pure-identity URI' that is meant to be used when reference is made to a particular physical object among information systems and business application software. Each coding scheme in the EPC identifier framework is discriminated by the use of a separate namespace. In the URI notation, this is indicated using a URI prefix such as *urn:epc:id:sgtin* or *urn:epc:id:sscc*. In the compact binary encoding of an EPC identifier, the namespace is instead indicated using a compact binary header, typically the first 8 bits of the binary encoding of an EPC identifier. The EPCglobal tag data standard provides details of the URI prefixes and corresponding binary header values. This namespace indicator (URI prefix or a compact binary header value) determines the length, type and structure of the EPC. Most EPCs include an element within their structure that corresponds to a serial number.

Version 1.4 of EPC supports the following alternative coding schemes: General Identifier (GID) GID-96; serialized versions of the GS1 Global Trade Item Number (GTIN), SGTIN-96 and SGTIN-198; GS1 Serial Shipping Container Code (SSCC), SSCC-96; GS1 Global Location Number (GLN), SGLN-96 and SGLN-195; GS1 Global Returnable Asset Identifier (GRAI), GRAI-96 and GRAI-170; GS1 Global Individual Asset Identifier (GIAI), GIAI-96 and GIAI-202; DOD Construct DoD-96; Global Service Relation

Number (GSRN), GSRN-96; and Global Document Type Identifier (GDTI), GDTI-96. Version 1.5 of the tag data standard additionally defines the formatting of the Tag Identifier (TID) memory bank and how to use Packed Objects for the formatting of additional data within the user memory bank.

Microsoft BizTalk RFID and Sun Java RFID support custom/private coding schemas (in addition to EPC), while SaviTrak uses both EPC and Savi Network's clientele coding scheme.

Ownership transfer of tags implies that the ownership of the supply chain, especially by retailers, has led to fears of tag owners potentially acquiring the capacity and the will to track tagged products even after they have left the supply chain. Since it is difficult to realize tag ownership transfer without a trusted third party (TTP), Kapoor, Zhou, and Piramuthu (2011) proposed a tag ownership sharing protocol and evaluate its properties.

The long expected ubiquitous deployment of RFID across the supply chain, especially by retailers, has led to fears of tag owners potentially acquiring the capacity and the will to track tagged products even after they have left the supply chain. Hence, an RFID middleware should take cognizance of these hypothetical consumer fears and the legal mandates that may apply to data collection in a given jurisdiction. If RFID communication protocols support special privacy enforcing attributes, the RFID middleware should seek to follow suit. For example, in the case of the kill command specified in the air interface protocol standard of EPC network, the middleware should provide a suitable kill password to the reader.

It will be desirable to have external sensors trigger readers to register events. A compelling reason for this would be the fact that in the majority of applications, readers do not need to operate continuously. Also, communication bandwidth limitations imply that readers should capture and transmit data only when events of interest occur, such as tags in a given class coming into a room. Moreover, for example, the Accada filtering middleware (Floerkemeier, Roduner, & Lampe, 2007) allows applications to define standing queries and subsequently subscribe to them. The subscription is utilized to configure the corresponding readers by means of the EPCglobal reader protocol. This allows the readers to more economically use the limited bandwidth. One potential use scenario of this service is to adapt the queries executed by a reader over the air interface, e.g., by reading a particular group of tags more frequently or only at certain time intervals. The RFID data filtering task could then be shared between the air interface and the event router. This publish-subscribe service is not reported explicitly in any of the architectures in Table 3, but it is indirectly possible in Microsoft BizTalk RFID via the device service provider interface layer and the RFID software development kit (SDK).

It is clear from Table 3 that, for the available commercial visibility systems, product data does not as yet include the processes, activities and the actors that the product experiences in its progress through the supply chain. This is desirable and is part of the PLIM model (Kiritsis, 2011; PROMISE, 2008) but the realization of PLIM at an industrial scale still seems far off the horizon. Since it is possible that in the long term most products will have EPC tags, of the four architectures compared in Table 3, EPC is the more likely to support PLIM in the future. However, since EPC was not designed primarily for PLIM, it would be necessary to modify and/or augment the architecture before it can fully support PLIM.

9. Conclusion

This paper has characterized visibility systems, technologies and architectures. It provides a new classification of visibility sys-

tems that extends some previous classifications. It then used survey by questionnaire to develop user needs and requirements for visibility systems. User needs and assessment criteria for visibility systems are industry-dependent. A clear picture that emerged from the development of the requirements for visibility systems is that the industry is more interested in simple systems that deliver immediate value to their customers than in complex concepts and systems. This is understandable: the industry exists to serve their customers and will not invest in what it perceives to be a technology whose potential will not be realized until someday in the future.

Some of the system requirements developed in this study were then used to assess four of the numerous existing architectures and models of supply chain visibility. The comparison of the four systems was not intended to judge the value of any of the systems but to identify the essential features and the differences between them, so as to exploit the potential for fine tuning the systems to special scientific and industry needs and for developing interoperability schemes between the systems.

Since there are several commercial and research/academic visibility systems, the paper conducted an extensive literature and provides in Table 2 an encompassing list of citations where recent architectures have been developed or can be found. Each source cited in Table 2 represents one system/architecture. Researchers, vendors and application developers will find Table 2 practical in the initial stages of their research and development efforts.

Research into applications of RFID and sensor networks to supply chain integration remains an active and fertile area of activity. Identification, location, navigation and geospatial data are nowadays at the core of technologies deployed to improve the efficiency and security of global trade. The data can be combined with reliable wireless shipment monitoring to provide shippers, logistics service providers, and terminal operators with easily accessible, accurate and actionable information. The information is used as a source of supply chain intelligence to identify and attend to both operational and systemic, longer term, issues. It is to be expected that the role of real-time inventory information on the supply chain's competitiveness will grow in the coming years. Technology is an enabler of both business efficiency and responsiveness.

This paper is based on an extensive review of the latest literature, but of course it is possible that some existing visibility systems and models have not come to the notice of the authors and have therefore not been mentioned in this paper.

9.1. Future research directions

The research reported in this paper will in the future examine the architectural frameworks of Section 7 from a more practical perspective, by actually implementing some of them and comparing their performance in terms of cost, resource requirements, user needs, suitability to small businesses, and customer service levels. In the remainder of this section we highlight some possible future directions beyond what we plan to pursue on our part.

RFID and sensor network devices are only some of dissimilar sources of data for real-time business intelligence. Other sources of data include ontologies (source ontology, user ontology, application ontology, shared ontologies, etc.) for modeling data sources and applications. In the near term we are likely to see more use of ontologies for supply chain intelligence. The future will also witness more research directed at using the real-time data collected at the frontline for directing the supply chain. Currently, a lot of information goes up to the upper echelons of the supply chain, but it is not sufficiently used in the reverse direction to direct the supply chain in the sense of effective adaptive control. One constructive model that we are developing in our University is a service that traces the progress of a product through the supply chain and

transmits this information to the various enterprise resource planning (ERP) systems of the supply chain partners, as well as pass instructions from the ERP systems back to the product itself in a purely dynamic and adaptive fashion. This is a key aspect of the event-driven controlling of supply chains. Other forms of adaptive control of supply chains have been proposed by Hong, Kim and Kim (2010), Hong, Yoo, Ko and Kim (2008), Ko, Kwak, Cho, and Kim (2011) and Yoo, Hong, and Kim (2009). Another direction for future research is a numerical/practical comparison of the architectures. Of course, such a project will require a sizeable financial resource and extensive infrastructure; it requires a collaborative effort by several investigators.

Since, in the meantime, item-level tagging will remain expensive, depending on the value of the item, some attention is likely to be directed in the future at selective tagging of products, especially of returnables. Modeling issues associated with gaining supply chain-wide visibility by tagging only a few of the returnable assets will feature prominently in the near future.

The future will also see a significant rise in the use of RFID in the healthcare sector (Ngai, Poon, Suk, & Ng, 2009), particularly for patient care and for fighting counterfeit drugs. This paper has not featured visibility structures developed specifically for the healthcare sector.

Instead of continuously developing and introducing new visibility systems and structures, efforts should be directed at worldwide adoption of one of the existing systems and standards. Then if ever new standards become necessary, it should only be required to achieve interoperability between the existing and new ones. The EPC network was intended and developed as a global standard but new architectures that extend the functionality and sensor array of EPC network beyond RFID keep emerging. Commercial considerations and the drive for innovation are unlikely to encourage global standardization; if anything, they are more likely to exert influence in the reverse direction. Since in the meantime there is no guarantee that a global standard in system architecture will emerge, a useful area of research is the development of plug and play interoperability tools between the various structures.

Another area of fruitful research is product lifecycle information management (PLIM) (PROMISE, 2008), given the potential uses as illustrated in this paper of this technology. However, no sizeable practical deployment of PLIM is likely to happen before embedded devices with suitable computing and communications capacity become affordable. Currently, product tracking and tracing with the barcode and RFID does not include information about the processes, activities and the actor to which the product has been subjected. PLIM will make product, activity and actor tracking realizable in the future. Tracking and tracing processes, activities and actors, in addition to products, will also require significant computational and communication resources onboard micro devices. This aspect of product visibility will play a greater role within PLIM.

Acknowledgments

The authors would like to thank the anonymous reviewers of the paper for their constructive criticisms that significantly helped to improve the quality and flow of the paper.

References

- Aberdeen Group. (2011). Omnitrol's inventory visibility, tracking, and traceability journey: Connecting store and supply chain. Analyst insight. Available at: <http://www.omnitrol.com/sites/default/files/Connecting%20Store%20and%20Supply%20Chain.pdf>. Last accessed 10/01/2013.
- Agaidi. (2010). AGAIDI: People tracking at Helsinki airport. Available at: http://www.agaidi.com/wp-content/uploads/2011/10/Agaidi-airport-solution_webinar.pdf. Last accessed 10/2/2012.

- Ahn, H. J., Childerhouse, P., Vossen, G., & Lee, H. (2012). Rethinking XML-enabled agile supply chains. *International Journal of Information Management*, 32, 17–23.
- Airlines of America (2009). *ATA Spec 2000. Automated identification and data capture (AIDC)*. 1301 Pennsylvania Avenue, NW Suite 1100, Washington, DC 20004: A4A Publications Department.
- Akyuz, G. A., & Rehan, M. (2009). Requirements for forming an e-supply chain. *International Journal of Production Research*, 47, 3265–3287.
- Amador, C., & Edmond, J. P. (2010). Evaluation of sensor readability and thermal relevance for RFID temperature tracking. *Computers and Electronics in Agriculture*, 73, 84–90.
- Amaral, L. A., Hessel, F. P., Bezerra, E. A., Correa, J. C., Longhi, O. B., & Dias, T. F. O. (2011). eCloudRFID – a mobile software framework architecture for pervasive RFID-based applications. *Journal of Network and Computer Applications*, 34, 972–979.
- Atzori, L., Iera, A., & Morabito, G. (2010). The internet of things: A survey. *Computer Networks*, 54, 2787–2805.
- Azuara, G., Tornos, J. L., & Salazar, J. L. (2012). Improving RFID traceability systems with verifiable quality. *Industrial Management and Data Systems*, 112, 340–359.
- Bardaki, C., Kourouthanassis, P., & Pramatar, K. (2011). Modeling the information completeness of object tracking systems. *Journal of Strategic Information Systems*, 20(3), 268–282.
- Bechini, A., Cimino, M. G. C. A., Marcelloni, F., & Tomasi, A. (2008). Patterns and technologies for enabling supply chain traceability through collaborative e-business. *Information and Software Technology*, 50, 342–359.
- Bechini, A., Cimino, M. G. C. A., & Tomasi, A. (2005). Using ebXML for supply chain traceability. In M. Funabashi & A. Grzech (Eds.), *Challenges of expanding internet: E-commerce, e-business, and e-government*. IFIP International Federation for Information Processing.
- Bensky, A. (2008). *Wireless positioning technologies and applications*. Boston: Artech House.
- Bottani, E., Montanari, R., & Volpi, A. (2010). The impact of RFID and EPC network on the bullwhip effect in the Italian FMCG supply chain. *International Journal of Production Economics*, 124, 426–432.
- Bottani, E., & Rizzi, A. (2008). Economical assessment of the impact of RFID technology and EPC system on the fast-moving consumer goods supply chain. *International Journal of Production Economics*, 112, 548–569.
- Botts, M. (2007). OpenGIS sensor model language (SensorML) implementation specification, version 1.0.0, open geospatial consortium. Available at: <http://www.opengeospatial.org/standards/requests/31>, Last accessed 20/02/2013.
- Carabelea, E., & Boissier, O. (2004). Benefits and requirements of using multi-agent systems on smart devices. *Lecture Notes in Computer Science*, 2790, 1091–1098.
- Cassina, J., Taisch, M., Potter, D., & Parlikad, A. K. (2008). Development of PROMISE architecture and PDKM semantic object model. In *Proceedings of 14th International Conference on Concurrent Enterprising*, June 23–25, 2008, Lisbon, Portugal (pp. 669–676).
- Chappel, D. A. (2004). *Enterprise service bus*. Sebastopol: O'Reilly Media Inc.
- Chatziantoniou, D., Pramatar, K., & Sotiropoulos, Y. (2011). Supporting real-time supply chain decisions based on RFID data streams. *The Journal of Systems and Software*, 84, 700–710.
- Chen, J.-L., Chen, M.-C., Chen, C.-W., & Chang, Y.-C. (2007). Architecture design and performance evaluation of RFID object tracking systems. *Computer Communications*, 30, 2070–2086.
- Chen, J. C., Cheng, C.-H., & Huang, P. B. (2013). Supply chain management with lean production and RFID application: A case study. *Expert Systems with Applications*, 40, 3389–3397.
- Chen, Y., Hsu, C.-Y., Liu, L., & Yang, S. (2012). Constructing a nutrition diagnosis expert system. *Expert Systems with Applications*, 39, 2132–2156.
- Chen, H., Zhu, Y., Hu, K., & Ku, T. (2011). RFID network planning using a multi-swarm optimizer. *Journal of Network and Computer Applications*, 34, 888–901.
- Cimino, M. G. C. A., & Marcelloni, F. (2011). Autonomic tracing of production processes with mobile and agent-based computing. *Information Sciences*, 181, 935–953.
- EPCglobal, Inc. (2004). The EPC network: Overview of design, benefits and security. Chicago: EPCglobal Inc.
- EPCglobal, Inc. (2004). The EPC network and the global data synchronization network (GDSN): Understanding the Information and the Information Networks. Chicago: EPCglobal Inc.
- EPCglobal, Inc. (2006). Application level events (ALE) specification, version 1.0 (ratified standard September 2005). Available at: http://www.gs1.org/gsm/kc/epcglobal/ale/ale_1_0-standard-20050915.pdf. Last accessed 20/02/2013.
- EPCglobal, Inc. (2006). Reader protocol standard, version 1.1 (ratified standard June 2006). Available at: autoid.mit.edu/CS/files/11/download.aspx. Last accessed 20/02/2013.
- EPCglobal Inc. (2010). The EPCglobal architecture framework (version 1.4). Available at: http://www.gs1.org/gsm/kc/epcglobal/architecture/architecture_1_4-framework-20101215.pdf. Last accessed 10/01/2013.
- EPCglobal, Inc. (2010). Low level reader protocol (LLRP), version 1.1 (ratified standard October 2010). Available at: http://www.gs1.org/gsm/kc/epcglobal/llrp/llrp_1_1-standard-20101013.pdf. Last accessed 20/02/2013.
- FDA (2004). US food and drug administration regulation 21CFR820, title 21 (Food and Drug Administration), Subchapter H (medical devices), part 820 (quality system regulation). Available at: <http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?CFRPart=820>. Last accessed 20/03/2013.
- Ferrer, G., Dew, N., & Apte, U. (2010). When is RFID right for you? *International Journal of Production Economics*, 124, 414–425.
- Floerkemeier, C., Roduner, C., & Lampe, M. (2007). RFID application development with the Accada middleware platform. *IEEE System Journal*, 1(2), 82–94.
- Food Standards Agency, UK (2002a). Guidance on the implementation of articles 11, 12 and 16 to 20 of regulation (EC) No. 178/2002 on general food law. Food Standards Agency of United Kingdom. Available at: http://ec.europa.eu/food/food/foodlaw/guidance/guidance_rev_7_en.pdf. Last accessed 28/03/2013.
- Food Standards Agency, UK (2002b). Traceability in the food chain: A preliminary study. Food Standards Agency of United Kingdom. Available at: <http://www.food.gov.uk/multimedia/pdfs/traceabilityinthefoodchain.pdf>. Last accessed 20/03/2013.
- Främling, K., Ala-Risku, T., Kärkkäinen, M., & Holmström, J. (2006). Agent-based model for managing composite product information. *Computers in Industry*, 57, 72–81.
- Främling, K., Ala-Risku, T., Karkkainen, M., & Holstrom, J. (2007). Design patterns for managing product life cycle information. *Communications of the ACM*, 50, 75–79.
- Främling, K., Harrison, M., Brusey, J., & Petrow, J. (2007). Requirements on unique identifiers for managing product lifecycle information: Comparison of alternative approaches. *International Journal of Computer Integrated Manufacturing*, 20, 715–726.
- Gibbs, B., & Damodaran, S. (2002). *ebXML concepts and application*. Indianapolis: Wiley Publishing Inc.
- Gomez, L., Laurent, M., & El Moustaine, E. (2012). Risk assessment along supply chain: A RFID and wireless sensor network integration approach. *Sensors & Transducers Journal*, 14, 269–282.
- Gunasekaran, A., & Ngai, E. W. T. (2004). Information systems in supply chain integration and management. *European Journal of Operational Research*, 159, 269–295.
- Gunasekaran, A., Ngai, E. W. T., & McGaughey, R. E. (2006). Information technology and systems justification: A review for research and applications. *European Journal of Operational Research*, 173, 957–983.
- Holmström, J., Kajosaari, R., Främling, K., & Langius, E. (2009). Roadmap to tracking based business and intelligent products. *Computer in Industry*, 60, 229–233.
- Hong, S. R., Yoo, J. S., Ko, J. M., & Kim, C. O. (2008). Direct neural network-based service level control in RFID-enabled supply chain. In *Proceedings of the 17th World Congress, International Federation of Automatic Control*, Seoul, Korea, July 6–11, 2008, pp. 4476–4480.
- Hong, S. R., Kim, S. T., & Kim, C. O. (2010). Neural network controller with online inventory feedback data in RFID-enabled supply chain. *International Journal of Production Research*, 48, 2613–2632.
- Hribernik, K. A., Rabe, L., Thoben, K.-D., & Schumacher, J. (2006). The product avatar as a product-instance-centric information management concept. *International Journal of Product Lifecycle Management*, 1, 367–379.
- Hsu, C., Levermore, D. M., Carothers, C., & Babin, G. (2007). Enterprise collaboration: On-demand information exchange using enterprise databases, wireless sensor networks, and RFID systems. *IEEE Transactions on Systems, Man, and Cybernetics – Part A: Systems and Humans*, 37, 519–532.
- Hsu, H.-H., & Liao, H.-T. (2011). A mobile RFID-based tour system with instant microblogging. *Journal of Computer and System Sciences*, 77, 720–727.
- Hsu, C., & Wallace, W. A. (2007). An industrial network flow information integration model for supply chain management and intelligent transportation. *Enterprise Information Systems*, 1, 327–351.
- Huang, H.-P., & Chang, Y.-T. (2011). Optimal layout and deployment for RFID systems. *Advanced Engineering Informatics*, 25, 4–10.
- Jakkhupan, W., Arch-int, S., & Li, Y. (2011). Business process analysis and simulation for the RFID and EPCglobal Network enabled supply chain: A proof-of-concept approach. *Journal of Network and Computer Applications*, 34, 949–957.
- Jedermann, R., & Lang, W. (2008). The benefits of embedded intelligence: Tasks and applications for ubiquitous computing in logistics. *The Internet of Things, Lecture Notes in Computer Science*, 4952, 105–122.
- Kapoor, G., Zhou, W., & Pirmuthu, S. (2011). Multi-tag and multi-owner RFID ownership transfer in supply chains. *Decision Support Systems*, 52(1), 258–270.
- Kärkkäinen, M., Ala-Risku, T., & Främling, K. (2004). Efficient tracking for short-term multi-company networks. *International Journal of Physical Distribution and Logistics Management*, 34(7), 545–564.
- Kärkkäinen, M., & Holmström, J. (2002). Wireless product identification: Enabler for handling efficiency, customization and information sharing. *Supply Chain Management*, 7(4), 242–252.
- Kärkkäinen, M., Holmström, J., Främling, K., & Artto, K. (2003). Intelligent products: A step towards a more effective project delivery chain. *Computers in Industry*, 50(2), 141–151.
- Kiritis, D. (2011). Closed-loop PLM for intelligent products in the era of the Internet of things. *Computer-Aided Design*, 43, 479–501.
- Ko, J. M., Kwak, C., Cho, Y., & Kim, C. O. (2011). Adaptive product tracking in RFID-enabled large-scale supply chain. *Expert Systems with Applications*, 38, 1583–1590.
- Kwok, S. K., Ting, J. S. L., Tsang, A. H. C., Lee, W. B., & Cheung, B. C. F. (2010). Design and development of a mobile EPC-RFID-based self-validation system (MESS) for production authentication. *Computers in Industry*, 61, 624–635.
- LaMarca, A., Chawathe, Y., Consolvo, S., Hightower, J., Smith, I., Scott, J., Sohn, T., Howard, J., Hughes, J., Potter, F., Tabert, J., Powlledge, P., Borriello, G., & Schilit, B. (2004). Place lab: Device positioning using radio beacons in the wild. Intel Research, Report No. IRS-TR-04-016.
- Lee, I., & Lee, B.-C. (2010). An investment evaluation of supply chain RFID technologies: A normative modeling approach. *International Journal of Production Economics*, 125, 313–323.

- Lee, I., & Lee, B.-C. (2012). Measuring the value of RFID investment: Focusing on RFID budget allocation. *IEEE Transactions on Engineering Management*, 59, 551–559.
- Lee, H., & Özer, Ö. (2007). Unlocking the value of RFID. *Production and Operations Management*, 16(1), 40–64.
- Liu, H., Darabi, H., Banerjee, P., & Liu, J. (2007). Survey of wireless indoor positioning techniques and systems. *IEEE Transactions on Systems, Man, and Cybernetics – Part C: Applications and Reviews*, 37(6), 1067–1080.
- MAFF Japan (2003). Guidelines for introduction of food traceability systems. Ministry of Agriculture, Forestry and Fisheries of Japan. Available at: <http://www.maff.go.jp/trace/guide_en.pdf>. Last accessed 20/03/2013.
- Martínez-Sala, A. S., Egea-López, E., García-Sánchez, F., & García-Haro, J. (2009). Tracking of returnable packaging and transport units with active RFID in the grocery supply chain. *Computers in Industry*, 60, 161–171.
- Mattoli, V., Mazzolai, B., Mondini, A., Zampolli, S., & Dario, P. (2010). Flexible tag dataloggers for food logistics. *Sensors and Actuators A*, 162, 316–323.
- Meyer, G. G., Främling, K., & Holmström, J. (2009). Intelligent products: A survey. *Computers in Industry*, 60, 137–148.
- Microsoft (2013). BizTalk RFID: Making RFID deployments easy, simple and economical. Available at: <http://msdn.microsoft.com/en-us/library/aa479354.aspx>. Last accessed 20/02/2013.
- MIMOSA (2006). MIMOSA's open system architecture for enterprise application integration (OSA-EAI) technical architecture summary. Available at: <http://www.mimosa.org/sites/default/files/TechDocs/OSA-EAI_Technical_Architecture_Summary_Dec_2006.pdf>. Last accessed 20/02/2013.
- Ngai, E. W. T., Poon, J. K. L., Suk, F. F. C., & Ng, C. C. (2009). Design of an RFID-based healthcare management system using information system design theory. *Information Systems Frontiers*, 11, 405–417.
- Norair, J. P. (2009). Introduction to DASH7 technologies. DASH7 Alliance. Available at: <https://dash7.memberclicks.net/assets/PDF/webinar%20111209%201.0.pdf>. Last accessed 20/01/2013.
- OATSystems (2006). OAT C4 architecture. www.oatsystems.com. Last accessed 20/02/2013.
- Odin (2013). Tag acquisition processor. <http://odinrfid.com/>. Last accessed 20/02/2013.
- Oliveira, R. R., Nogueira, C. F., Costa, A. C., Barbosa, L. J., & Prado, P. M. (2012). SWTRACK: An intelligent model for cargo tracking based on off-the-shelf mobile devices. *Expert Systems with Applications*, 40, 2023–2031.
- Omnitrol Networks (2010). Edge application and sensor engine. www.omnitrol.com. Last accessed 20/02/2013.
- Oztekin, A., Pajouh, F. M., Delen, D., & Swim, L. K. (2010). An RFID network design methodology for asset tracking in healthcare. *Decision Support Systems*, 49, 100–109.
- Papetti, P., Costa, C., Antonucci, F., Figorilli, S., Solaini, S., & Menesatti, P. (2012). A RFID web-based infotracng system for the artisanal Italian cheese quality traceability. *Food Control*, 27, 234–241.
- PearTrack (2013). Available at: <http://www.peartrack.com/>. Last accessed 2013/03/22.
- Prabhu, B. S., Su, X., Ramamurthy, H., Chu, C.-C., & Gadhi, R. (2005). Win-RFID – a middleware for the enablement of radio frequency identification based applications. In R. Shorey & C. M. Choon (Eds.), *Mobile, wireless and sensor networks: Technology, applications and future directions*. New York: Wiley.
- PROMISE (2008). PROMISE architecture series. Architecture references, volumes 1–5 (1: Overview, 2: Core PAC interface, 3: Messaging interface and data services, 4: PDKM system object model, 5: Decision support system). Available at: <http://cl2m.com/wiki/114>. Last accessed 20/02/2013.
- Ranasinghe, D. C., Harrison, M., Främling, K., & McFarlane, D. (2011). Enabling through life product-instance management: Solutions and challenges. *Journal of Network & Computer Applications*, 34, 1015–1031.
- Römer, K., Schoch, T., Mattern, F., & Dübendorfer, T. (2004). Smart identification frameworks for ubiquitous computing applications. *Wireless Networks*, 10(6), 689–700.
- Rönkkö, M., Kärkkäinen, M., & Holmström, J. (2007). Benefits of an item-centric enterprise-data model in logistics services: A case study. *Computers in Industry*, 58, 814–822.
- Sarac, A., Absi, N., & Dauzère-Pérez, S. (2010). A literature review on the impact of RFID technologies on supply chain management. *International Journal of Production Economics*, 128, 77–95.
- Savi Networks (2013). www.savinetworks.com. Last accessed 2013/03/22.
- Shafiei, F., Sundaram, D., & Piramuthu, S. (2012). Multi-enterprise collaborative decision support system. *Expert Systems with Applications*, 39, 7637–7651.
- Shih, D. H., Sun, P. L., & Lin, B. (2005). Securing industry-wide EPC network with WS-security. *Industrial Management and Data Systems*, 105(7), 972–996.
- Shin, T.-H., Chin, S., Yoon, S.-W., & Kwon, S.-W. (2011). A service-oriented integrated information framework for RFID/WSN-based intelligent construction supply chain management. *Automation in Construction*, 20, 706–715.
- Souza, R. D., Goh, M., Subdarakani, B., Wai, W. T., Toh, K., & Yong, W. (2011). Return on investor calculator for RFID ecosystem of high tech company. *Computers in Industry*, 62, 820–829.
- Spieß, P., Bornhövd, C., Lin, T., Haller, S., & Schaper, J. (2007). Going beyond auto-ID: A service-oriented smart items infrastructure. *Journal of Enterprise Information Management*, 20(3), 356–370.
- Sun Microsystems Inc. (2005). The Sun Java™ System RFID software architecture: A technical white paper. http://www.sun.com. Last accessed 28/03/2013.
- Sundaram, D., Zhou, W., Piramuthu, S., & Pienaar, S. (2010). Knowledge-based RFID enabled web service architecture for supply chain management. *Expert Systems with Applications*, 37, 7937–7946.
- Tajima, M. (2007). Strategic value of RFID in supply chain management. *Journal of Purchasing & Supply Management*, 13, 261–273.
- Tesoriero, R., Tebar, R., Gallud, J. A., Lozano, M. D., & Penichet, V. M. R. (2010). Improving location awareness in indoor spaces using RFID technology. *Expert Systems with Applications*, 37, 894–898.
- Thakur, M., & Hurburgh, C. R. (2009). Framework for implementing traceability system in the bulk grain supply chain. *Journal of Food Engineering*, 95, 617–626.
- Thiesse, F., & Fleisch, E. (2008). On the value of location information to lot scheduling in complex manufacturing processes. *International Journal of Production Economics*, 112, 532–547.
- Trienekens, J. H., Wognum, P. M., van der Beulens, A. J. M., & Vorst, J. G. A. J. (2012). Transparency in complex dynamic food supply chains. *Advanced Engineering Informatics*, 26, 55–65.
- Tse, Y. K., & Tan, K. H. (2012). Managing product quality risk and visibility in multi-layer supply chain. *International Journal of Production Economics*, 139, 49–57.
- Véronneau, S., & Roy, J. (2009). RFID benefits, costs, and possibilities: The economical analysis of RFID deployment in a cruise corporation global service supply chain. *International Journal of Production Economics*, 122, 692–702.
- Vrba, P., Macúrek, F., & Mařík, V. (2008). Using radio frequency identification in agent-based control systems for industrial applications. *Engineering Applications of Artificial Intelligence*, 21, 331–342.
- Wang, L., Ting, J. S. L., & Ip, W. H. (2013). Design of supply chain pedigree interactive dynamic explore (SPIDER) for food safety and implementation of hazard analysis and critical control points (HACCPs). *Computers and Electronics in Agriculture*, 90, 14–23.
- White, A., Daniel, E. M., & Mohdzain, M. (2005). The role of emergent information technologies and systems in enabling supply chain agility. *International Journal of Information Management*, 25, 396–410.
- Woo, S. H., Choi, Y. J., Kwak, C., & Kim, O. C. (2009). An active product state tracking architecture in logistics sensor networks. *Computers in Industry*, 60, 149–160.
- Xia, D., & Chen, B. (2011). A comprehensive decision-making model for risk management of supply chain. *Expert Systems with Applications*, 38, 4957–4966.
- Xu, L. D. (2011). Information architecture for supply chain quality management. *International Journal of Production Research*, 49, 183–198.
- Xu, D., Wijesooriya, C., Wang, Y.-G., & Beydoun, G. (2011). Outbound logistics exception monitoring: A multi-perspective ontologies' approach with intelligent agents. *Expert Systems With Applications*, 38(11), 13604–13611.
- Yoo, J. S., Hong, S. R., & Kim, C. O. (2009). Service level management of nonstationary supply chain using direct neural network controller. *Expert Systems with Applications*, 36, 3574–3586.
- Zhou, W. (2009). RFID and item-level information visibility. *European Journal of Operational Research*, 198, 252–258.